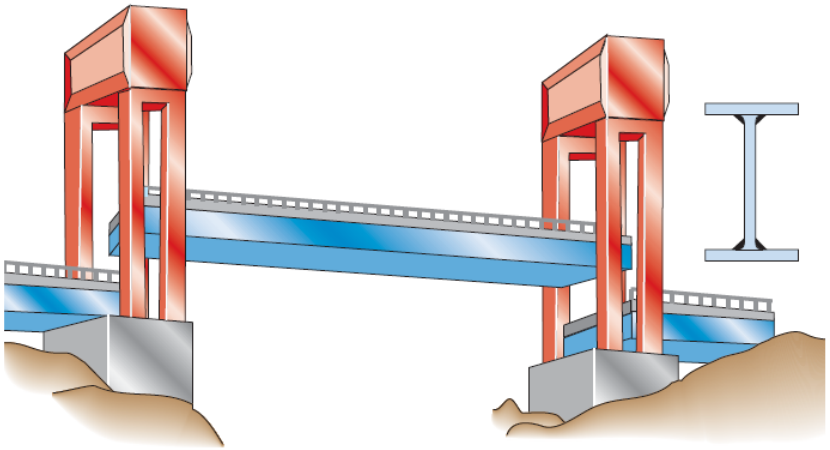
Bending Internal Force

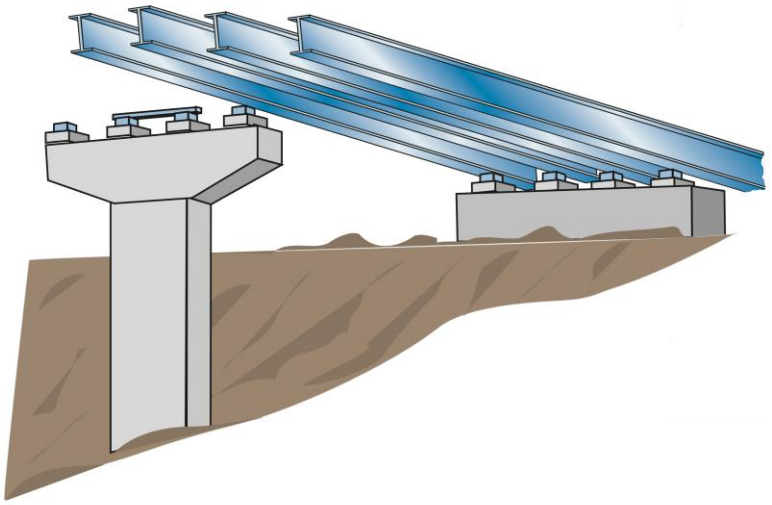
Note: content marked in * is more advanced knowledge

Introduction





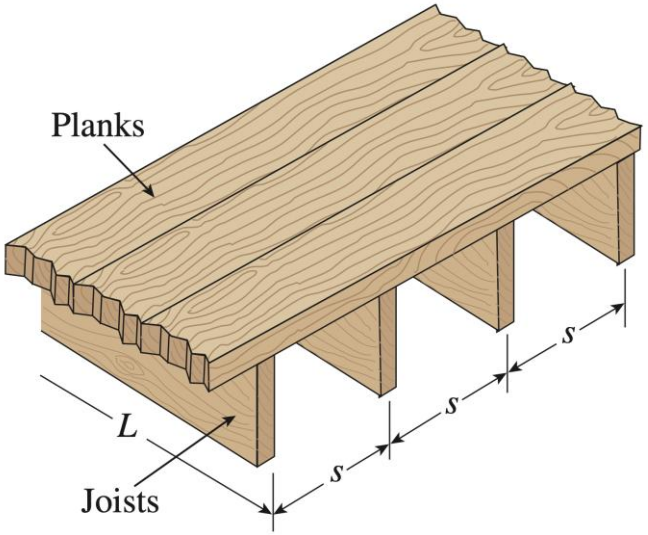




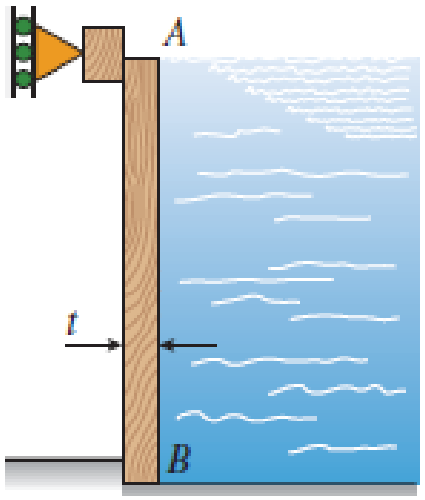
Introduction

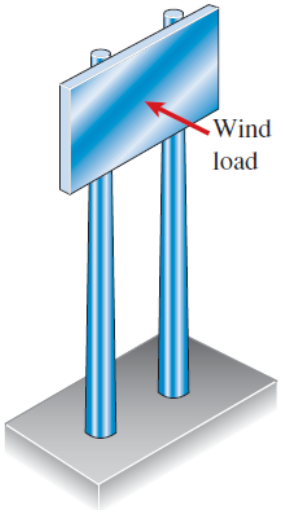




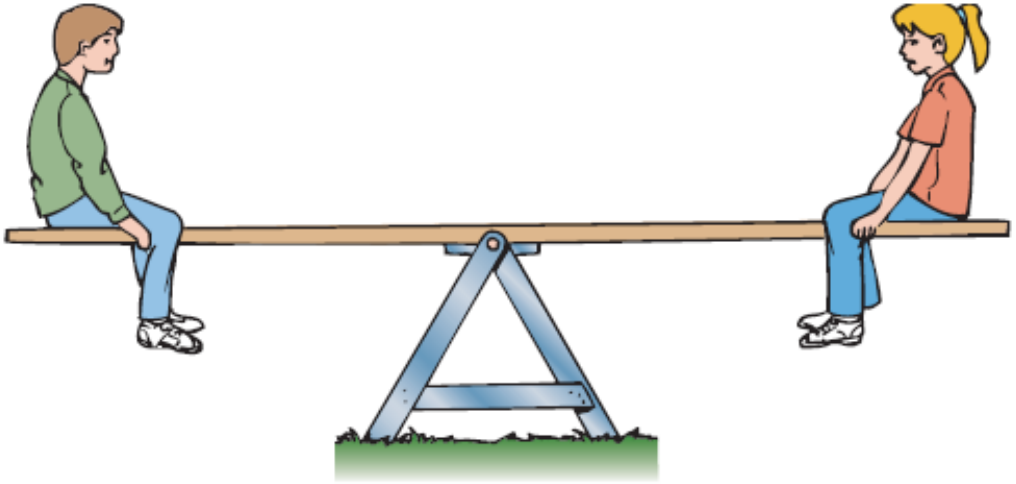


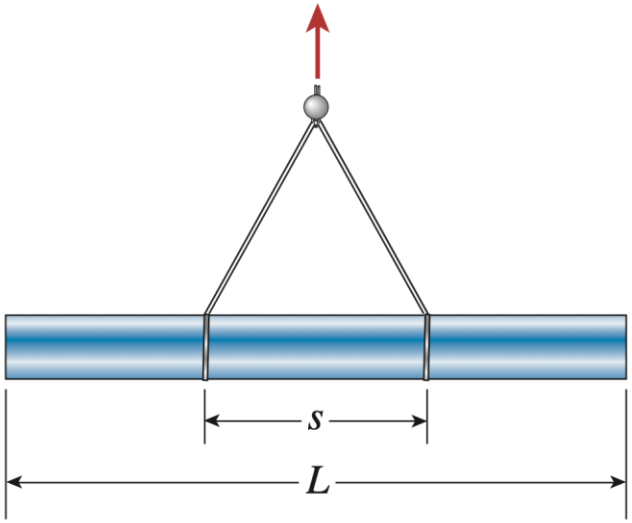


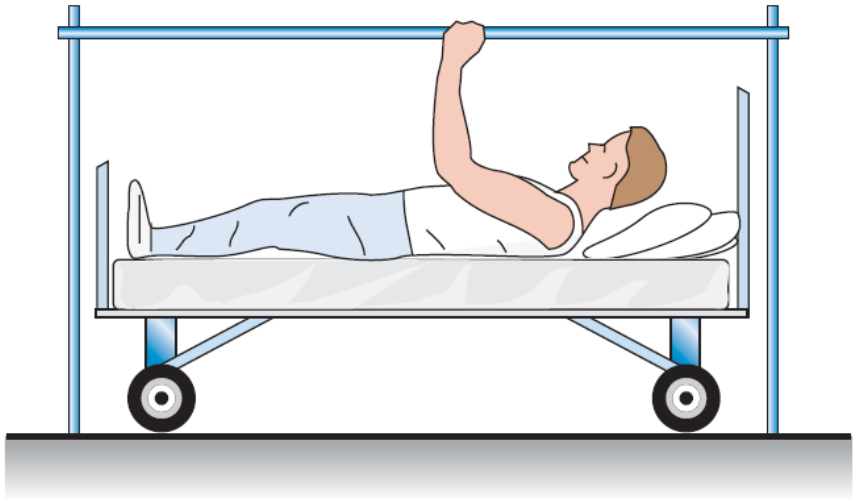


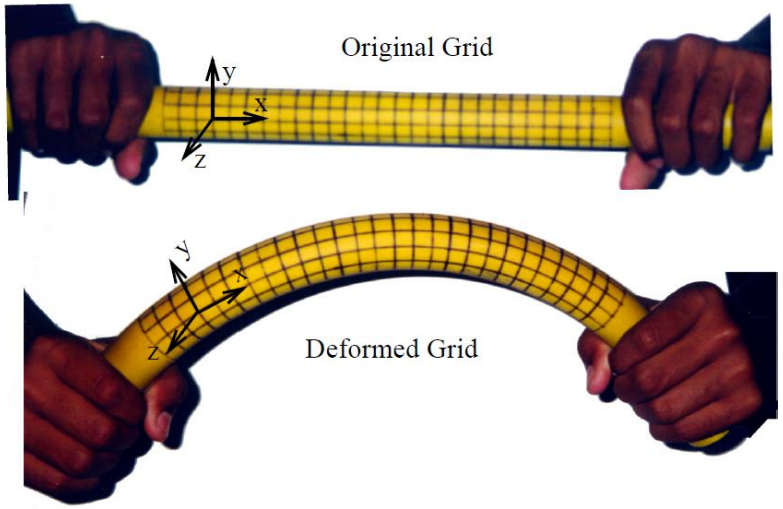


Introduction

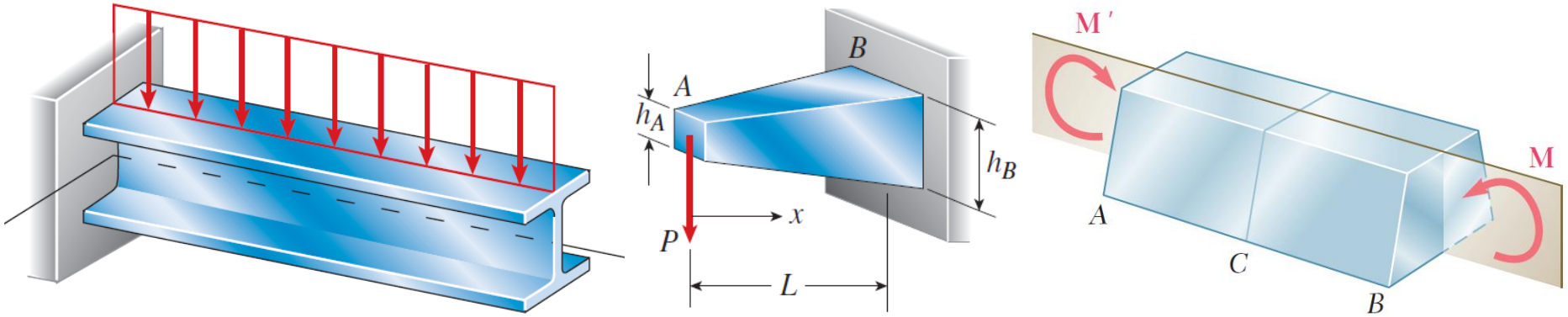








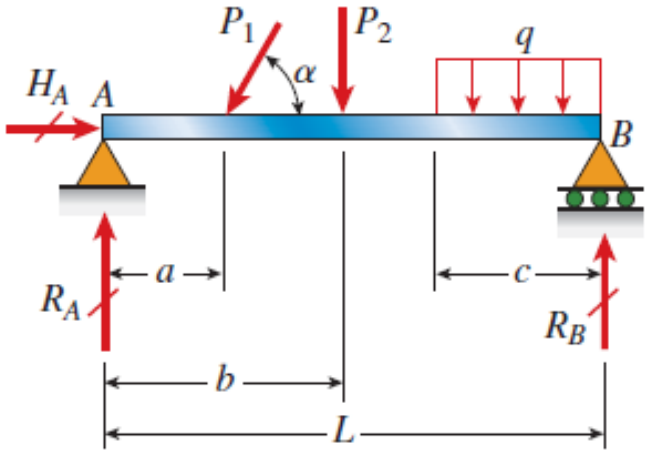
Symmetric Bending



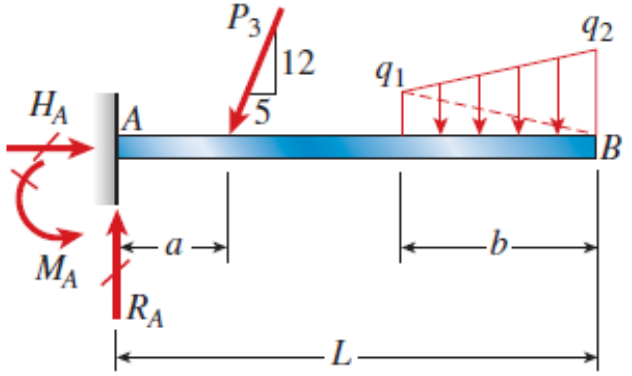
- **Beam:** bar members subjected to lateral loads, that is, forces or moments having their vectors perpendicular to the axis of the bar.

- **Symmetric bending:** all external forces & moments acting in the symmetric longitudinal plane (beam axis being bent from a straight line to a curve within the symmetric longitudinal plane).

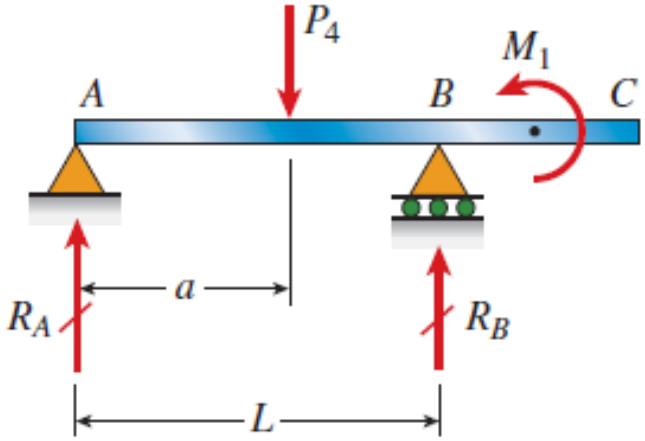
Types of Beams (All are statically determinate beams)



Simply supported beam

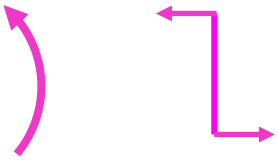


Cantilever beam



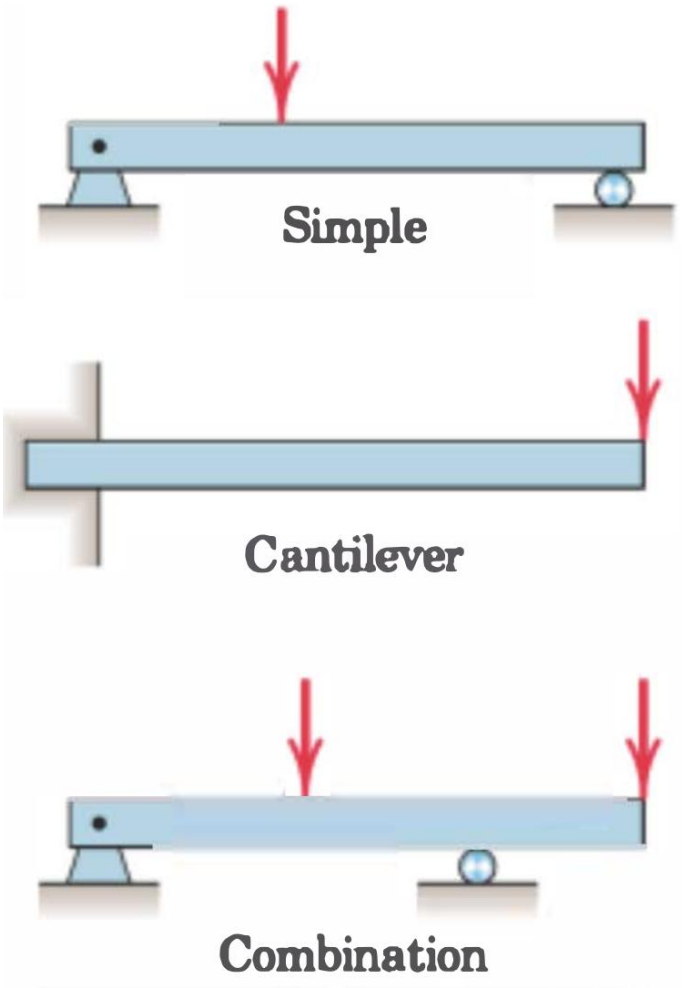
Beam with an overhang

Concentrated moment (force couple)

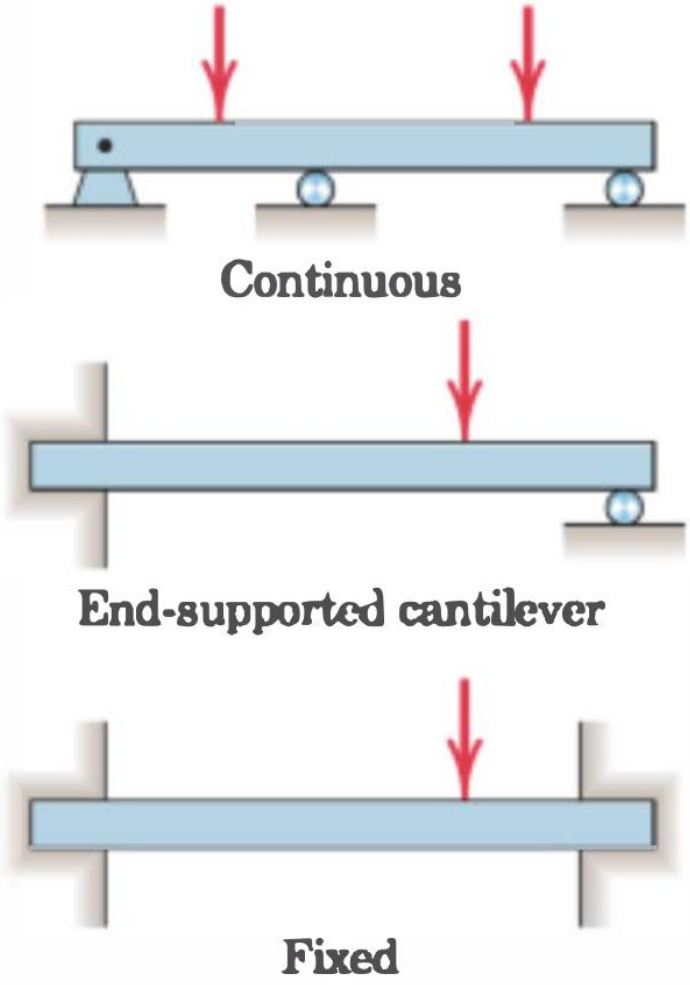


Determinacy of Beams

- Statically determinate beams

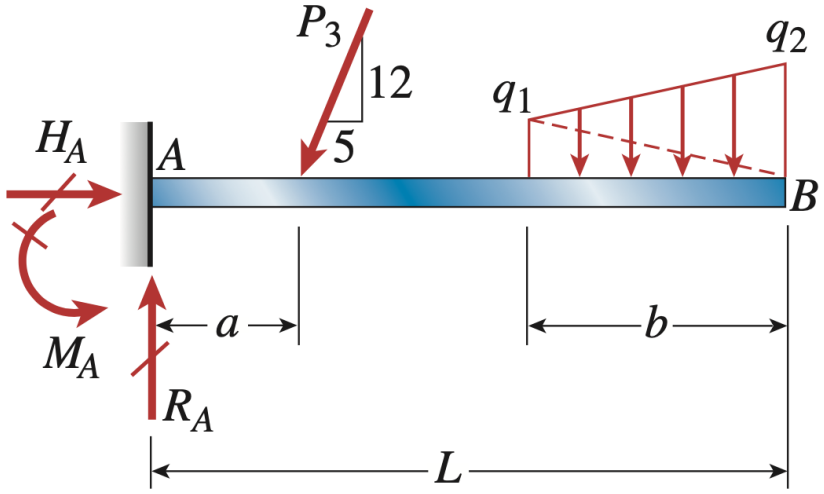


- Statically indeterminate beams



Criteria: if the reaction forces can be calculated using equilibrium equations **alone**

Statically determinate beams – Finding reaction forces



Equations of equilibrium

$$\sum F_X = 0 \qquad \sum F_Y = 0 \qquad \sum M = 0$$

(1) Equilibrium of forces in the horizontal direction gives $H_A = \frac{5P_3}{13}$

(2) Equilibrium in the vertical direction give $R_A = \frac{12P_3}{13} + \left(\frac{q_1 + q_2}{2}\right)b$

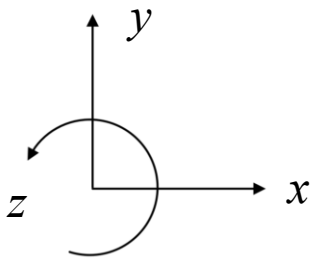
(3) Moment equilibrium

$$\Sigma M_A = 0 \quad M_A - \left(\frac{12P_3}{13}\right)a - \frac{q_1 b}{2}\left(L - \frac{2b}{3}\right) - \frac{q_2 b}{2}\left(L - \frac{b}{3}\right) = 0$$

Moment arm about A

from which

$$M_A = \frac{12P_3 a}{13} + \frac{q_1 b}{2}\left(L - \frac{2b}{3}\right) + \frac{q_2 b}{2}\left(L - \frac{b}{3}\right)$$



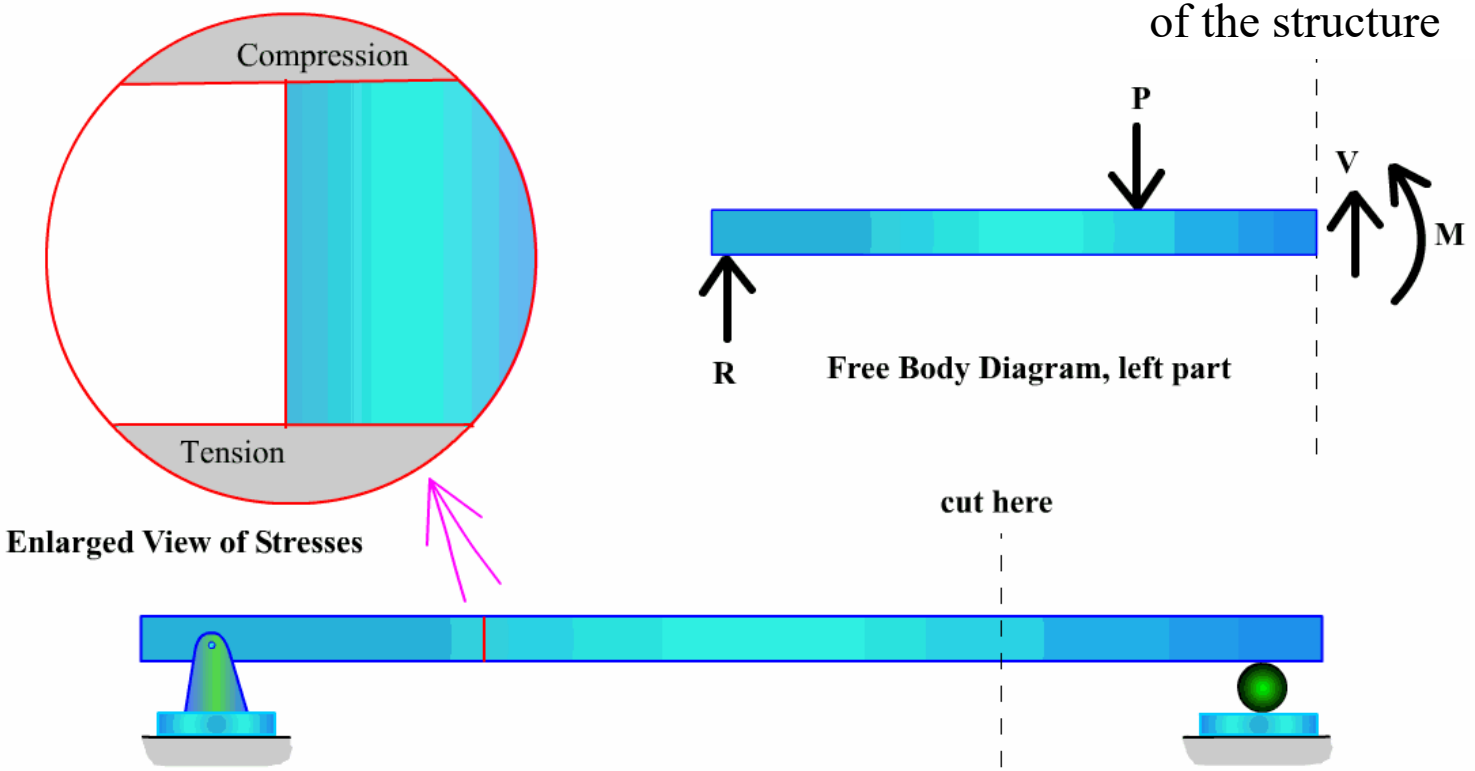
Right-handed Cartesian coordinate system

(The positive x and y axes point right and up, and the positive z axis points outward.)

Internal Forces in Beams

Internal force = resultants of stress at the cross section

The action to a proportion of the structure by the rest of the structure

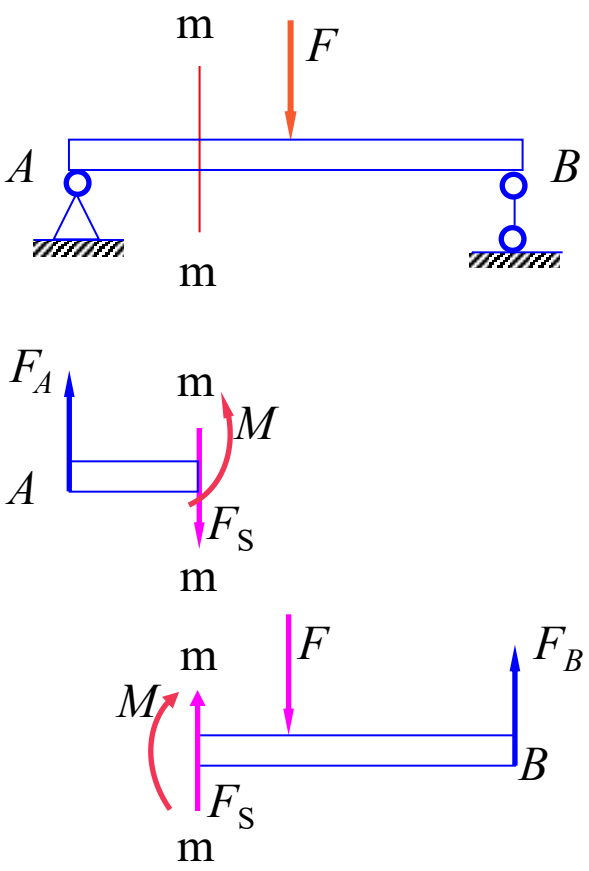


Members subjected to an external load applied perpendicularly to a longitudinal axis of the member.

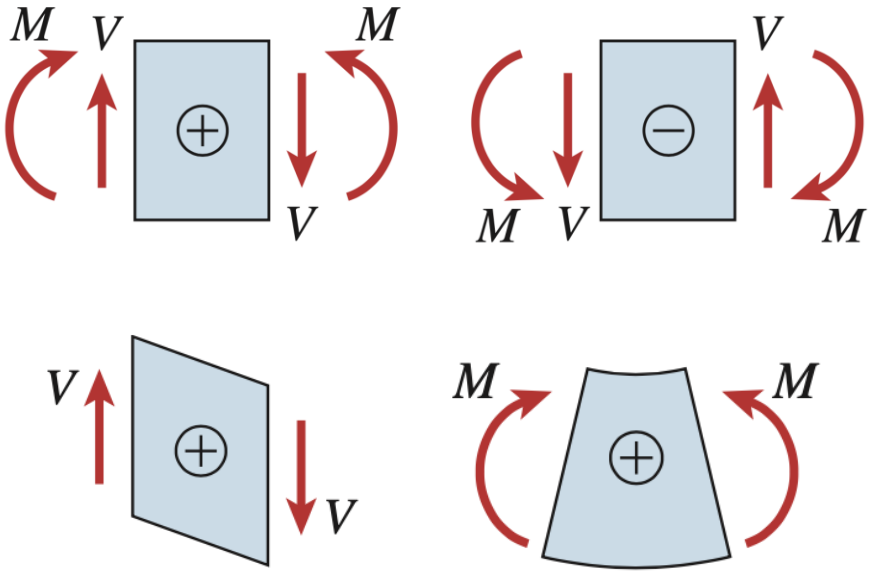
Internal Forces in Beams

- Internal forces on cross-sections of bending beams

method of sections



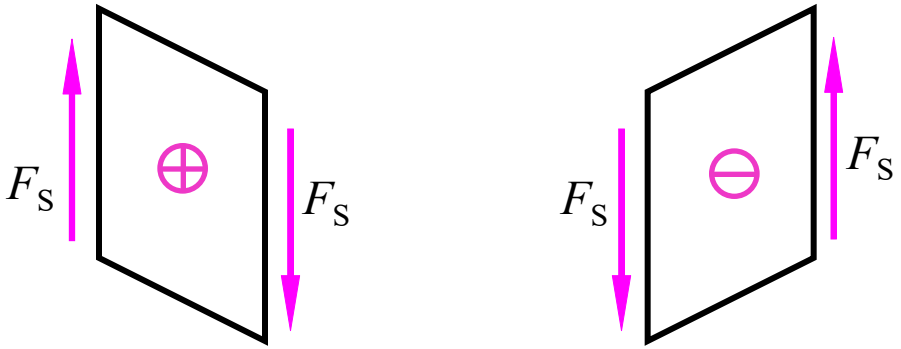
- The internal forces on cross-section m-m can be simplified as *shear force* and *bending moment* acting at cross-sectional centroid.
- The directions are *reversed*, but all are *positive* internal force/moment



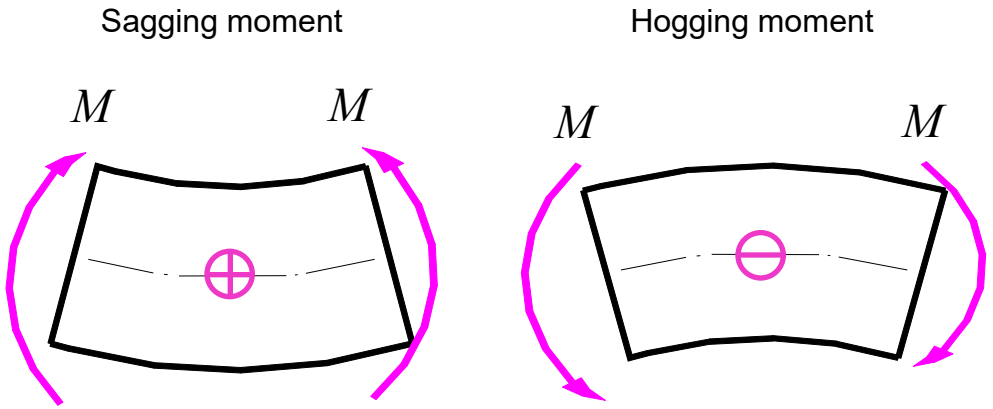
Deformations (highly exaggerated) of a beam element caused by (left) shear forces, and (right) bending moments

Sign Convention of Shear Force and Bending Moment (Internal)

- Shear force
  - Positive: left up / right down (clockwise loop against the material)



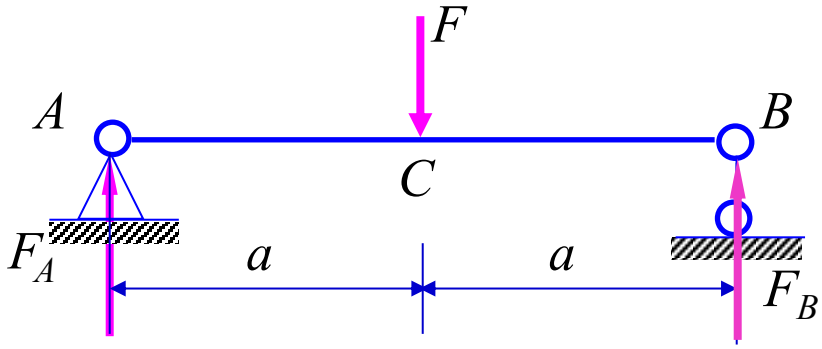
- Bending moment
  - Positive: upper half under compression / lower half under tension (top concave / bottom convex)



This sign convention is DIFFERENT from the right-handed coordinate system.
(depends on how it deforms material, rather than by its direction in space)

Example 1

- Find the shear force and bending moment at the left and right cross-section at point *C*.



- Solution:

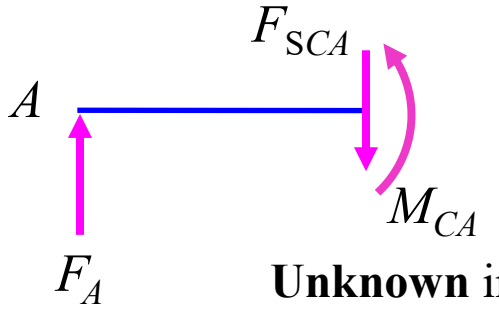
1. Reaction forces at the supports

$$F_A = F_B = \frac{F}{2}$$

2. Internal forces (assuming positive) at the left cross-section of *C*.

$$F_{SCA} = F_A = \frac{F}{2}$$

$$M_{CA} = F_A \cdot a = \frac{Fa}{2}$$

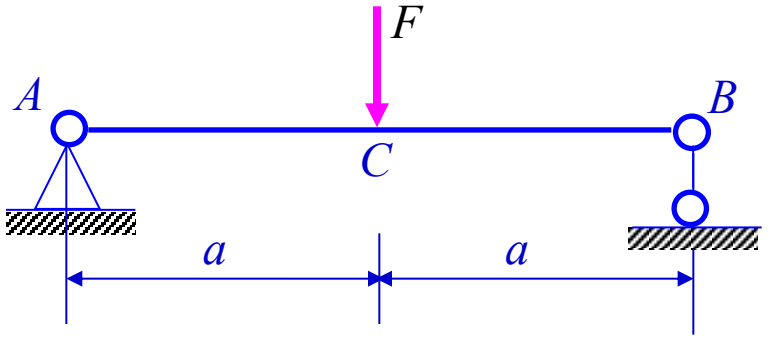


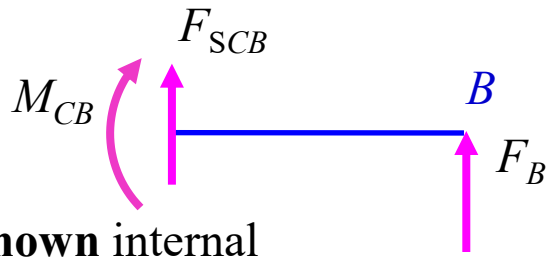
Unknown internal force/moment, shown in **positive**

3. Internal forces (assuming positive) at the right cross-section of *C*.

$$F_{SCB} = -F_B = -\frac{F}{2}$$

$$M_{CB} = F_B \cdot a = \frac{Fa}{2}$$



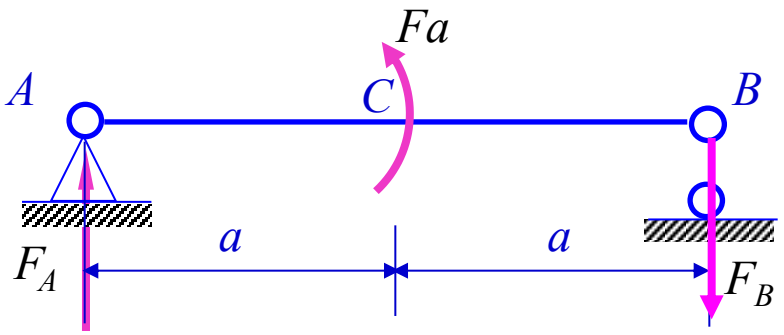


Unknown internal force/moment, shown in **positive**

Note: For a concentrated force acting at *C*, the change of bending moment is zero while that of shear force is equal to the magnitude of the concentrated force.

Example 2

- Find the shear force and bending moment at the left and right cross-section at point *C*.

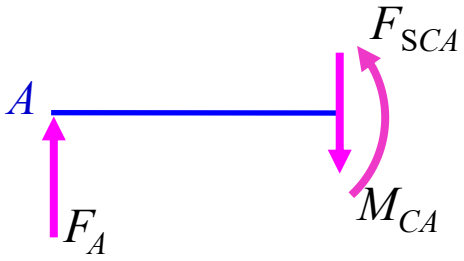


- Solution:

1. Reaction forces at the supports

$$F_A = F_B = \frac{F}{2}$$

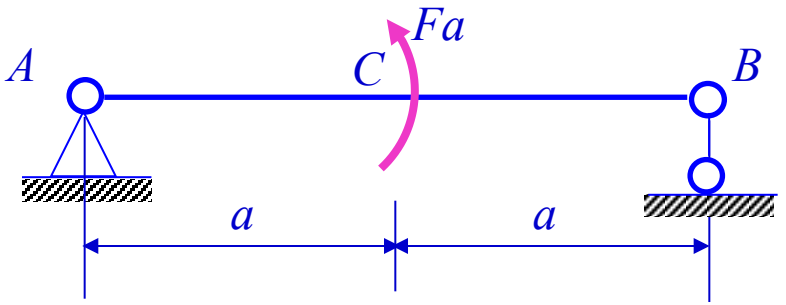
2. Internal force at the left cross-section of *C*.



$$F_{SCA} = F_A = \frac{F}{2}$$

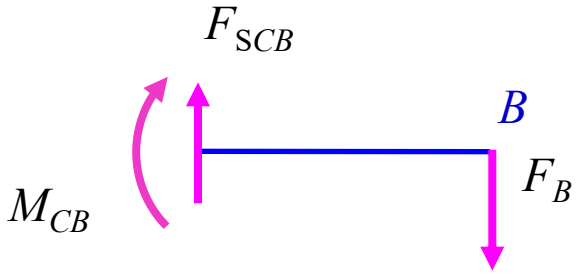
$$M_{CA} = F_A \cdot a = \frac{Fa}{2}$$

3. Internal force at the right cross-section of C.



$$F_{SCB} = F_B = \frac{F}{2}$$

$$M_{CB} = -F_B \cdot a = -\frac{Fa}{2}$$



Note: For a concentrated moment acting at *C*, the change of shear force is zero while that of bending moment is equal to the magnitude of the concentrated moment.

Direct Calculation of Shear Force

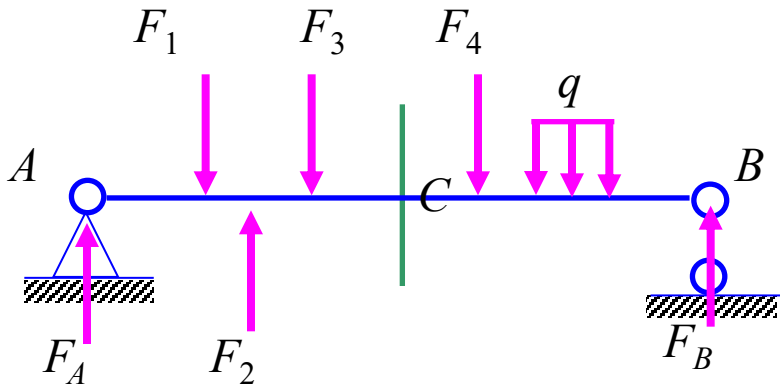
- The shear force at a beam cross-section is equal to the net of the external forces collected from either side of the cross-section.

- Upward net from left portion or downward net from right portion results in **positive** shear force at the cross-section.

- All forces acting on a beam, including the reaction forces at the supports, must be taken into consideration.

Example 3

- Directly calculate the shear force at cross-section C without the explicit use of method of section.



- Solution:
  - Based on the left portion from cross-section C:

$$F_{SC} = F_A - F_1 + F_2 - F_3$$

  - Based on the right portion from cross-section C:

$$F_{SC} = -F_B + ql + F_4$$

Note: due to equilibrium, the same answer would be obtained.

Direct Calculation of Bending Moment

- The bending moment at a beam cross-section is equal to the net of the external moments with respect to the cross-section centroid, collected from either side of the cross-section

- Clockwise net from left portion or counter-clockwise net from right portion results in **positive** bending moment.

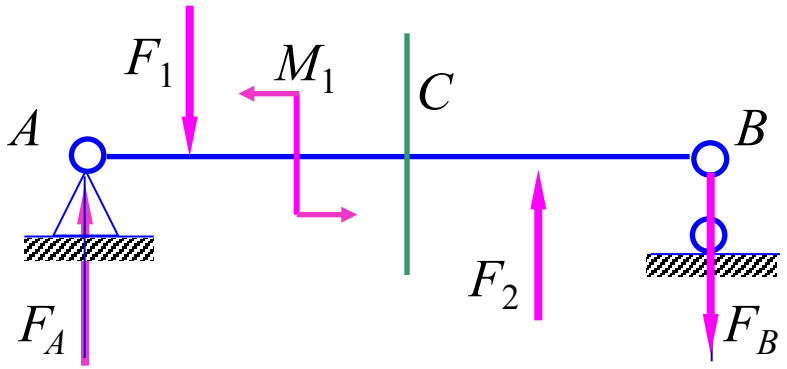
- All forces acting on a beam, including the reaction forces at the supports, must be taken into consideration.

Example 4

- Directly calculate the bending moment at cross-section C without the explicit use of method of section.

- Solution:

  - Based on the left portion from cross-section C



$$M_C = F_A d_A - F_1 d_1 - M_1$$

  - Based on the right portion from cross-section C

$$M_C = -F_B d_B + F_2 d_2$$

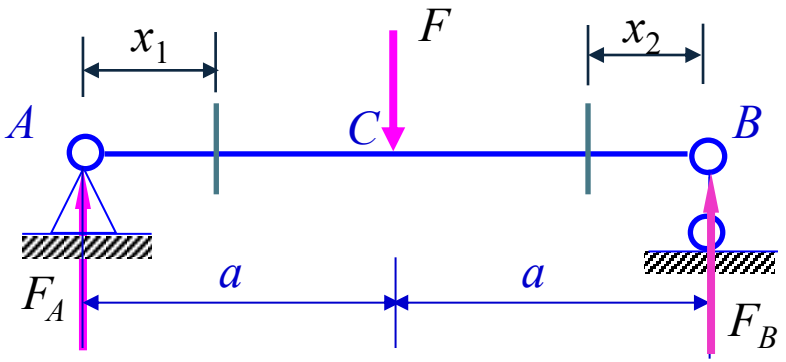
Note: due to equilibrium, the same answer would be obtained.

Diagram of Shear Forces & Bending Moments

- Equation of shear forces and bending moments
  - Equation of shear forces: $F_S = F_S(x)$
  - Equation of bending moments: $M = M(x)$
  - x: denotes the position of cross-section.
- Diagram of shear forces and bending moments
  - Abscissa (x-axis coordinate): cross-section position (x)
  - Ordinate (y-axis coordinate): shear forces (F_S) / bending moments (M)

Example 5

- Draw the diagram of shear forces and bending moments



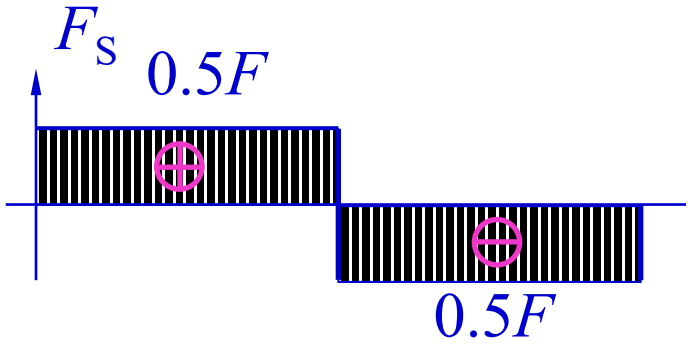
- Solution:

1. Reaction force at the supports: $\boldsymbol{F_A = F_B = \frac{F}{2}}$

2. Equation of internal forces

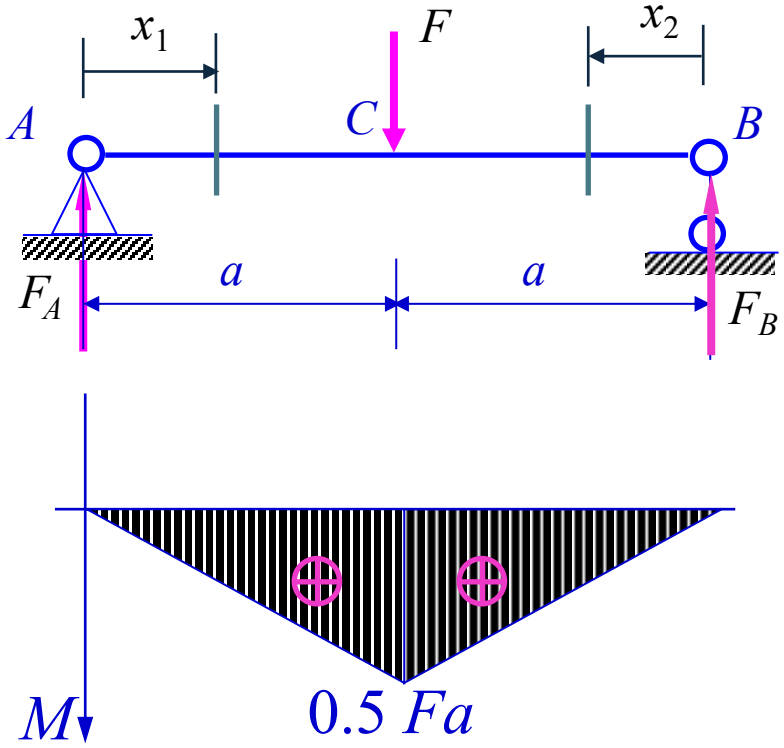
$$F_S(x_1) = F_A = \frac{F}{2} \qquad \boldsymbol{x_1 \in [0, a)}$$

$$F_S(x_2) = -F_B = -\frac{F}{2} \qquad \boldsymbol{x_2 \in [0, a)}$$



$$M(x_1) = F_A x_1 = \frac{Fx_1}{2} \quad x_1 \in [0, a)$$

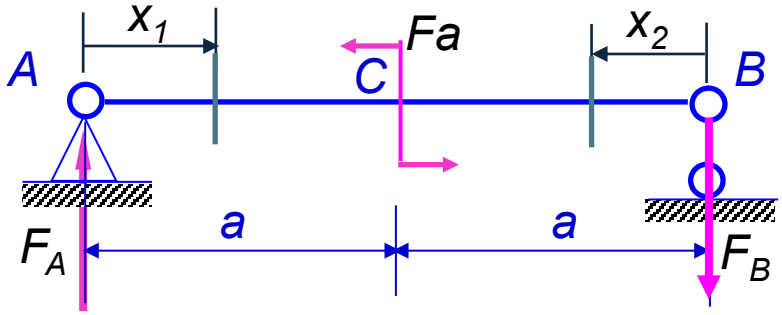
$$M(x_2) = F_B x_2 = \frac{Fx_2}{2} \quad x_2 \in [0, a)$$



By convention, bending moment is plotted on the tension side.

Example 6

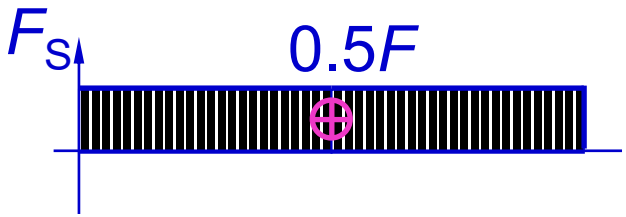
- Draw the diagram of shear forces and bending moments



- Solution:

1. Reaction forces at the supports

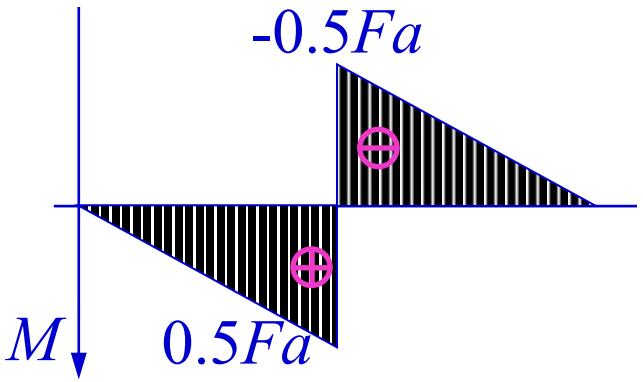
$$F_A = F_B = \frac{F}{2}$$



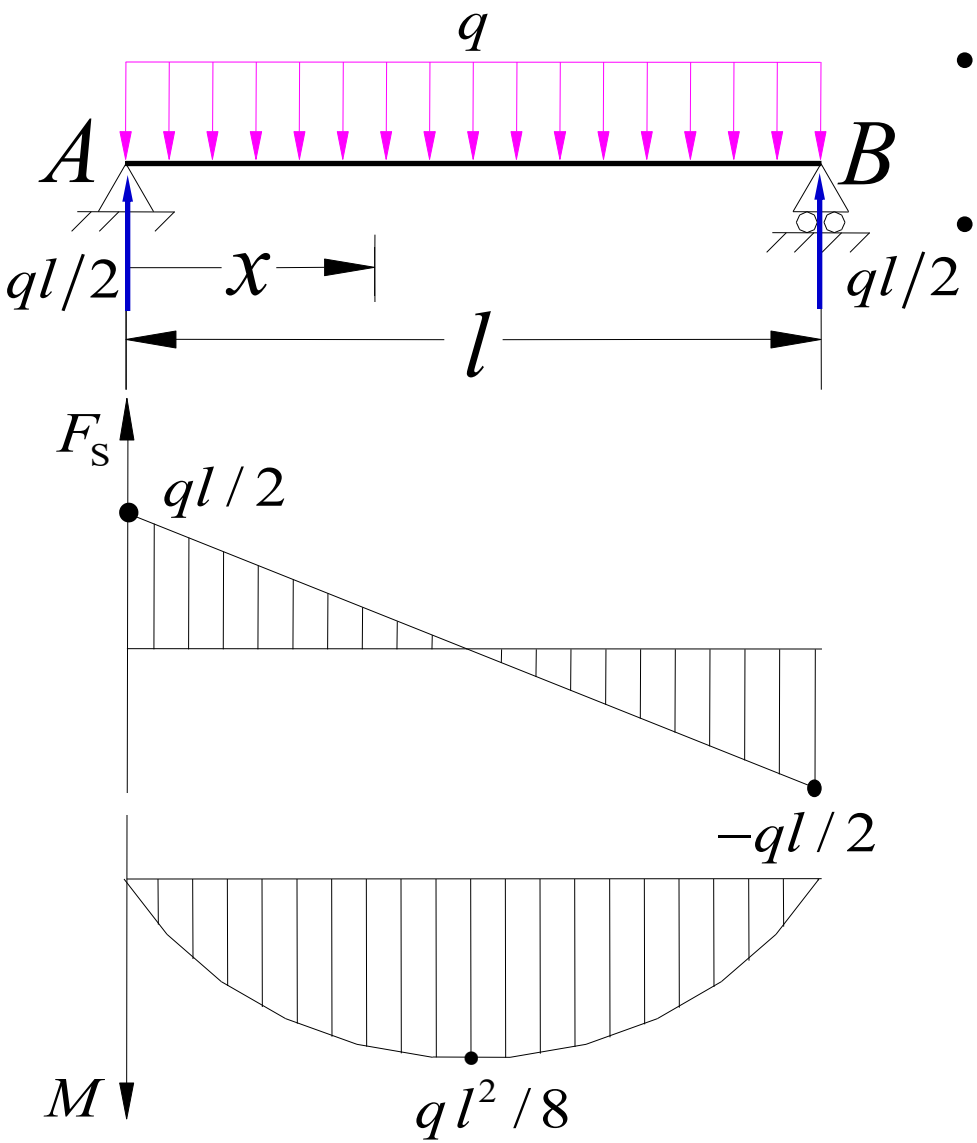
2. Equations of internal forces

$$F_S(x_1) = \frac{F}{2};\quad F_S(x_2) = \frac{F}{2}$$

$$M(x_1) = \frac{Fx_1}{2};\quad M(x_2) = -\frac{Fx_2}{2}$$



Example 7



3. Diagram of internal forces

- Plot the diagram of shear forces and bending moments
- Solution:

1. Reaction forces at the supports

2. Equation of internal forces

$$F_S(x) = \frac{q\,l}{2} - qx$$

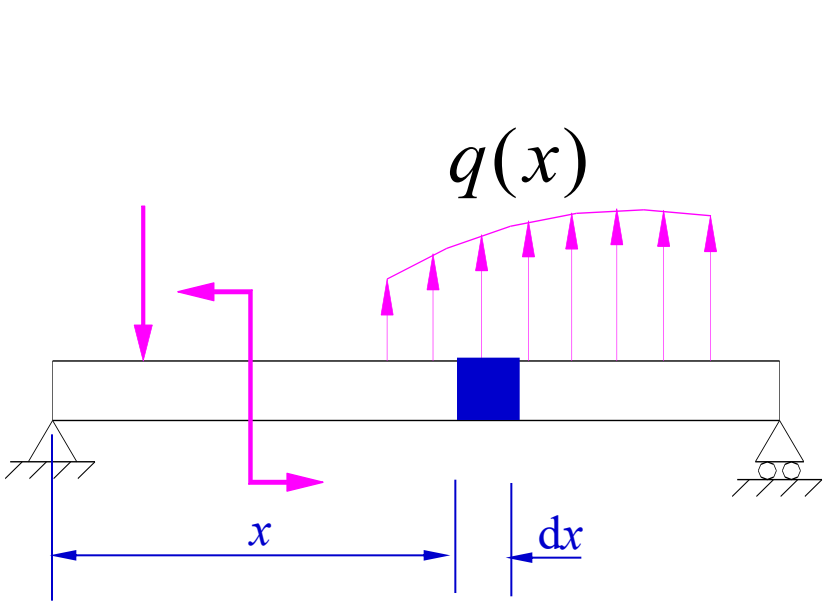
$$M(x) = \frac{q\,l}{2}x - qx\frac{x}{2}$$

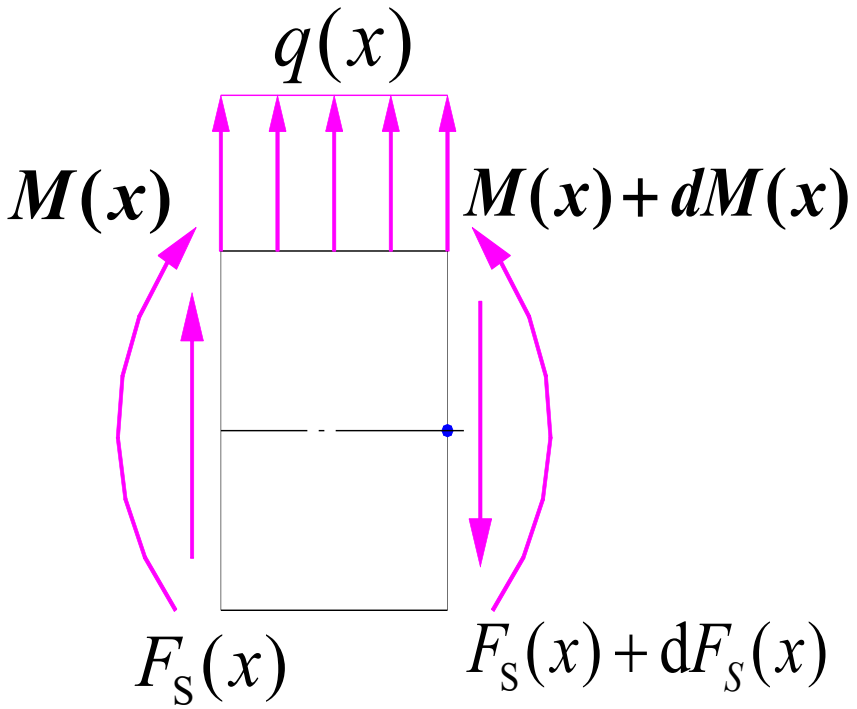
$$= -\frac{q}{2}\left(x - \frac{l}{2}\right)^2 + \frac{q\,l^2}{8}$$

$$(0 \le x \le l)$$

Relations among Loads, Shear Forces and Bending Moments

(Sign convention of $q(x)$: up positive / down negative.)





$$F_S(x) + q(x)\,\mathrm{d}x - F_S(x) - \mathrm{d}F_S(x) = 0$$

$$M(x) + \mathrm{d}M(x) - \frac{1}{2}q(x)\,\mathrm{d}^2x - M(x) - F_S(x)\,dx = 0$$

Relations among Loads, Shear Forces and Bending Moments

$$F_S(x) + q(x)\,\mathrm{d}x - F_S(x) - \mathrm{d}F_S(x) = 0$$

$$\Rightarrow \quad q(x) = \frac{dF_S(x)}{dx}, \qquad F_{S2} - F_{S1} = \int_{x_1}^{x_2} q(x)\,dx$$

$$M(x) + \mathrm{d}M(x) - \frac{1}{2}q(x)\,\mathrm{d}^2x - M(x) - F_S(x)\,dx = 0$$

$$\Rightarrow \quad F_S(x) = \frac{\mathrm{d}M(x)}{\mathrm{d}x}, \qquad M_2 - M_1 = \int_{x_1}^{x_2} F_S(x)\,dx$$

$$\Rightarrow \quad q(x) = \frac{\mathrm{d}M^2(x)}{\mathrm{d}x^2}$$

Sign convention of $q(x)$: up positive / down negative.

Relations among Loads, Shear Forces and Bending Moments

- Without distributed load: $q(x) = 0$

$$q(x) = \frac{dF_S(x)}{dx} = 0 \quad \Rightarrow \quad F_S(x) = \text{const}$$

Note: the slope of the diagram of shear forces is zero.

$$\frac{dM(x)}{dx} = F_S(x) = const \quad \Rightarrow \quad M(x) = (\text{const}) \times x + (\text{const})$$

Note: the slope of the diagram of bending moments is constant.

Relations among Loads, Shear Forces and Bending Moments

- With uniformly distributed load: $q(x) = \text{const}$

$$q(x) = \frac{dF_S(x)}{dx} = \text{const} \quad \Rightarrow \quad F_S(x) = (\text{const}) \times x + (\text{const})$$

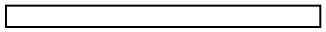
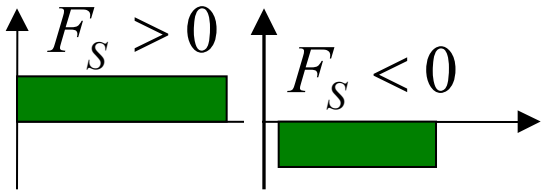
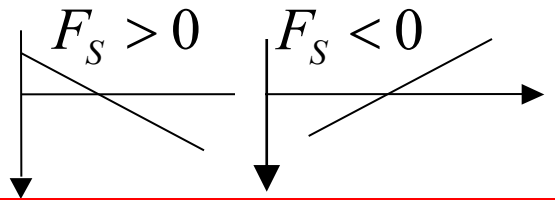
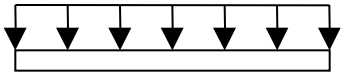
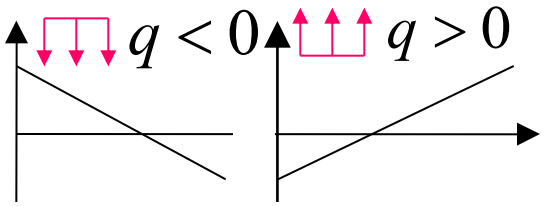
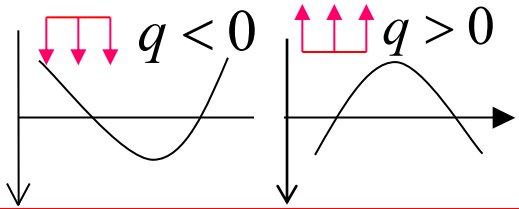
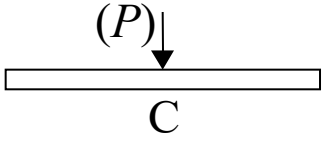
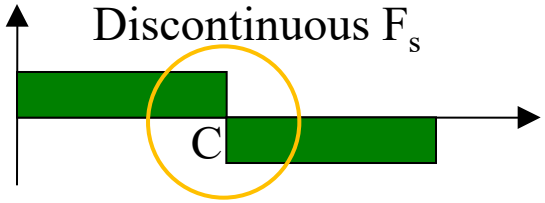
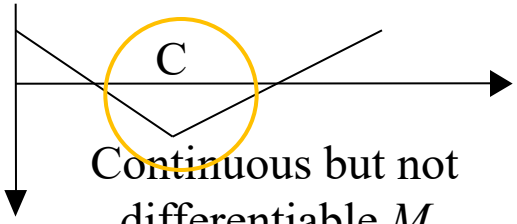
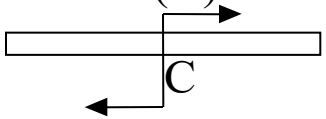
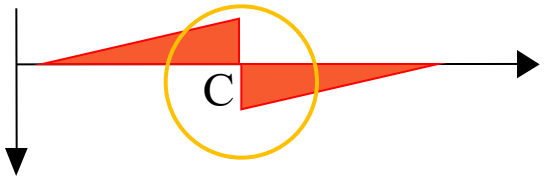
Note: the slope of the diagram of shear forces is constant.

$$\frac{dM(x)}{dx} = F_S(x) \quad \Rightarrow$$

$$M(x) = (\text{const}) \times x^2 + (\text{const}) \times x + (\text{const})$$

Note: $M(x)$ is a second order polynomial of x.

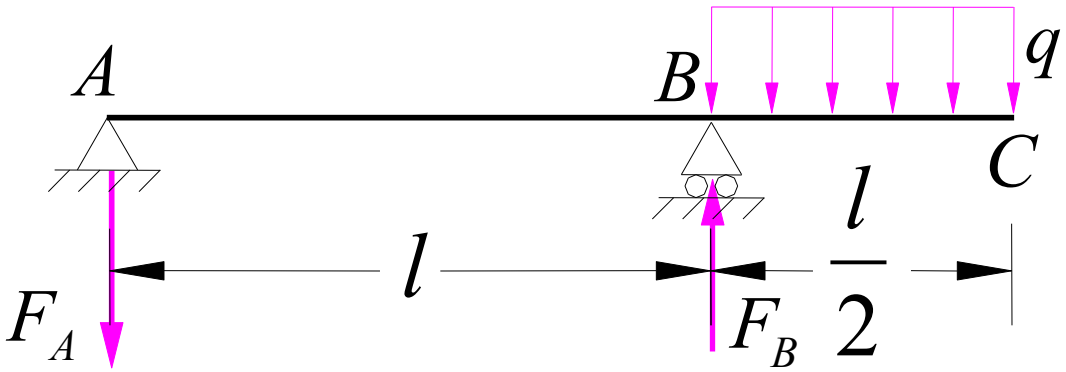
Relations among Loads, Shear Forces and Bending Moments

Loads	Diagram of shear forces	Diagram of bending moments
No load (P=0 & q=0) 		
Uniformly distributed q=const 		
Concentrated load (P) 		 Continuous but not differentiable M
Concentrated moment (m) 		

Example 8

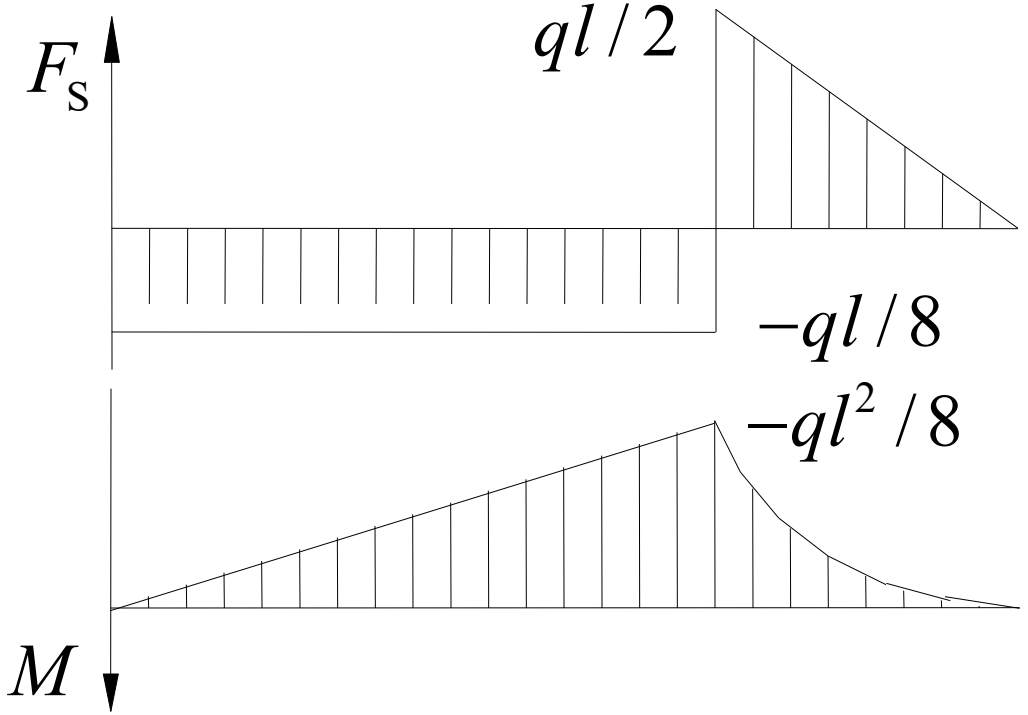
- Draw the diagram of shear forces and bending moments.

- Solution:



$$F_A = \frac{ql}{8}$$

$$F_B = \frac{5ql}{8}$$

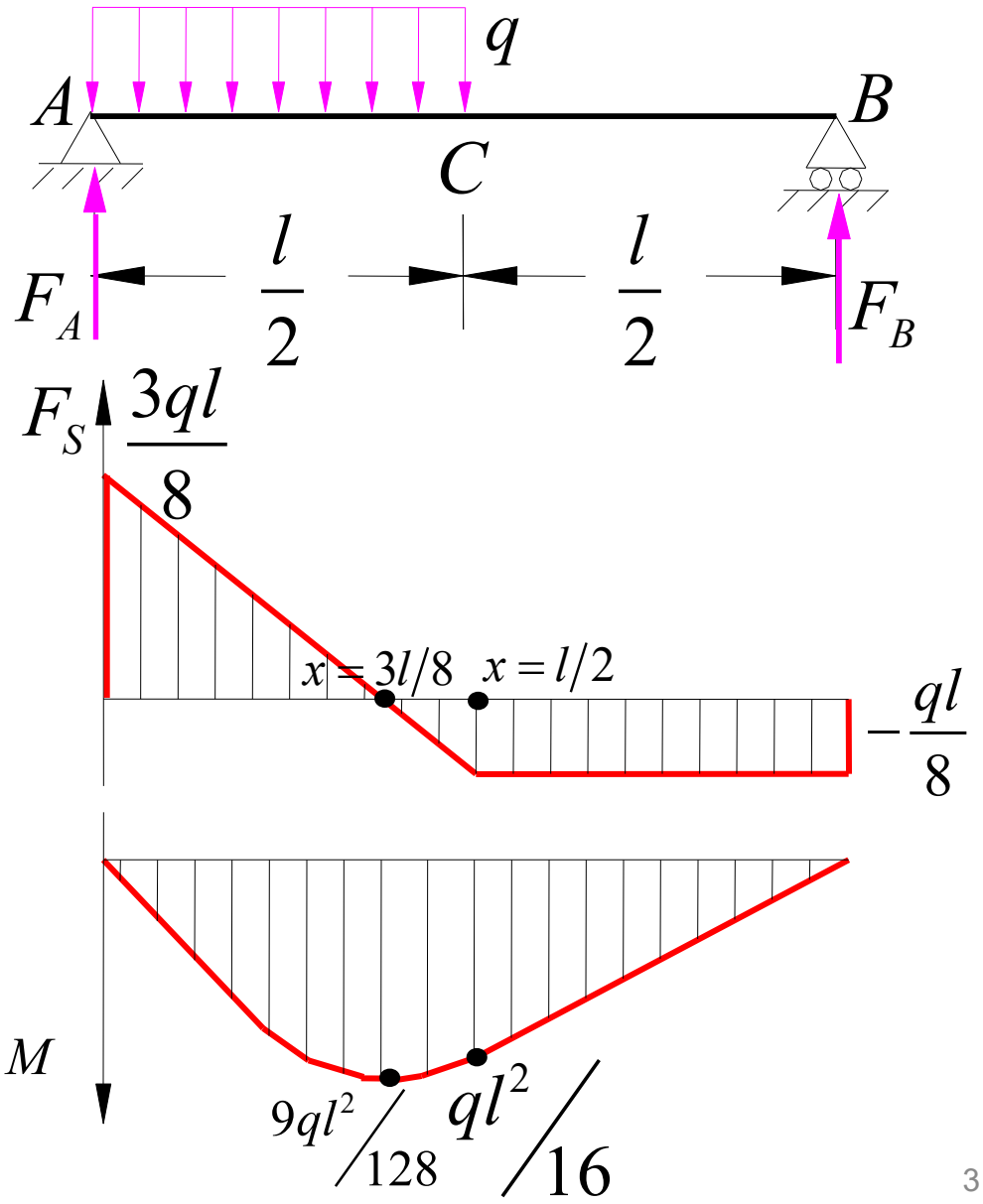


Example 9

- Draw the diagram of shear forces and bending moments.
- Solution:

$$F_A = \frac{3ql}{8}$$

$$F_B = \frac{ql}{8}$$



Example 10

- Draw the diagram of shear forces and bending moments.
- Solution:

$$F_A = qa$$
$$F_B = qa$$

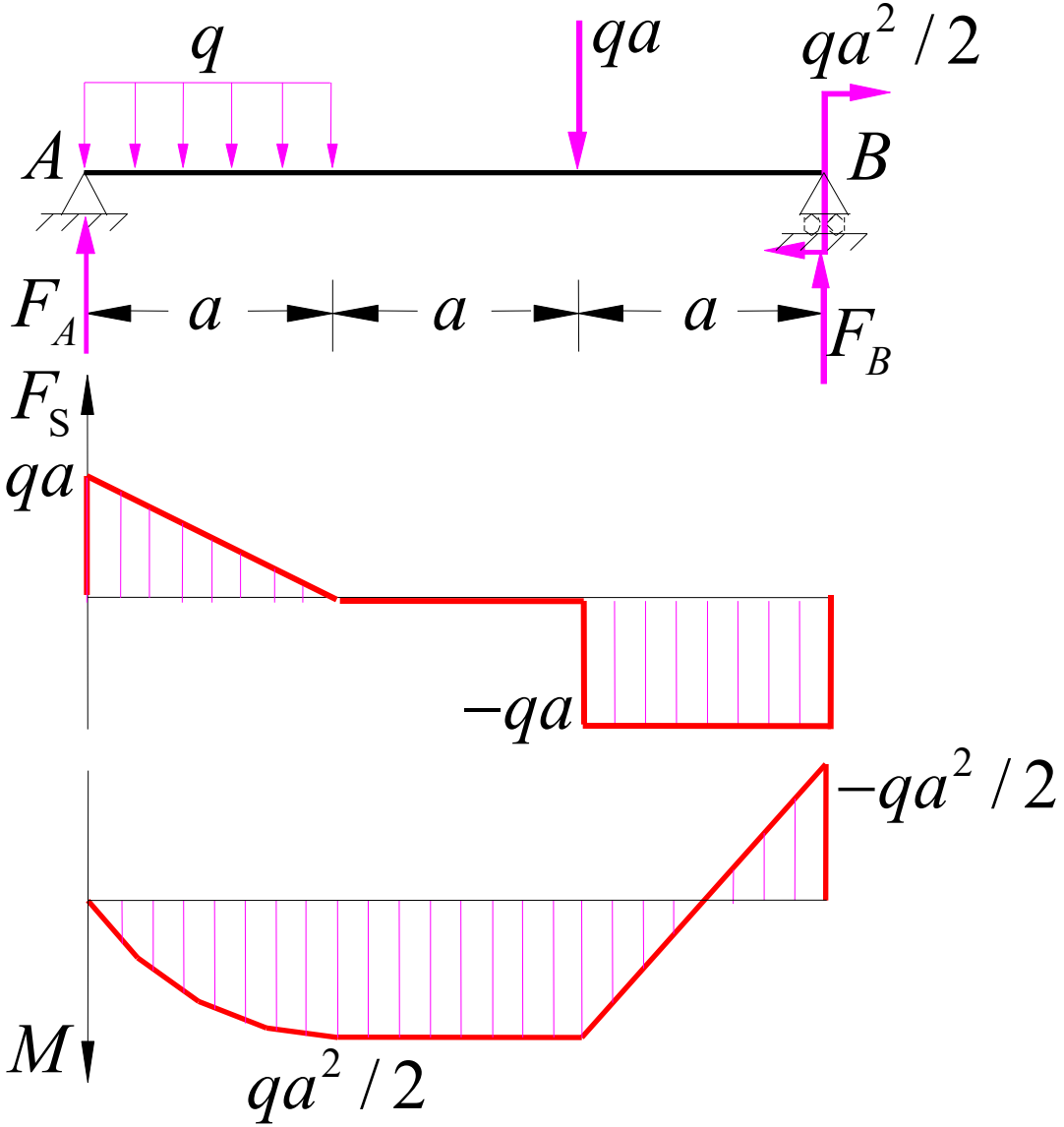


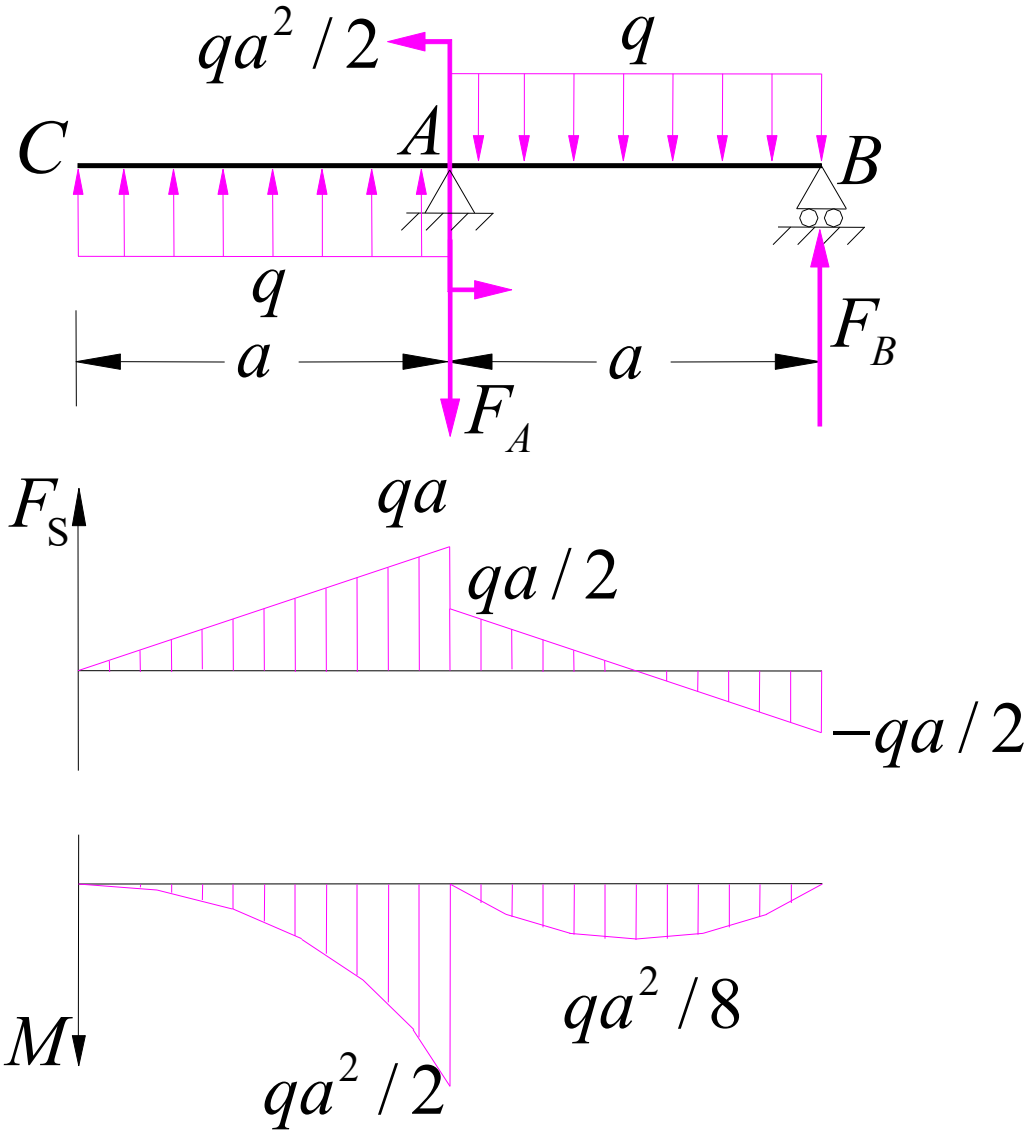
Diagram starts from 0 and returns to 0 (diagrams should always close)

Example 11

- Draw the diagram of shear forces and bending moments.
- Solution:

$$F_A = \frac{qa}{2}$$

$$F_B = \frac{qa}{2}$$



Analysis of Simple Frames*

Diagram of Internal Forces for Plane Frames

- Plane frames are composed of prismatic bars with different orientations in the same plane.
- Neighboring bars are connected via rigid joints
- Internal forces acting on cross-sections of component bars include axial force, shear force, and bending moment.

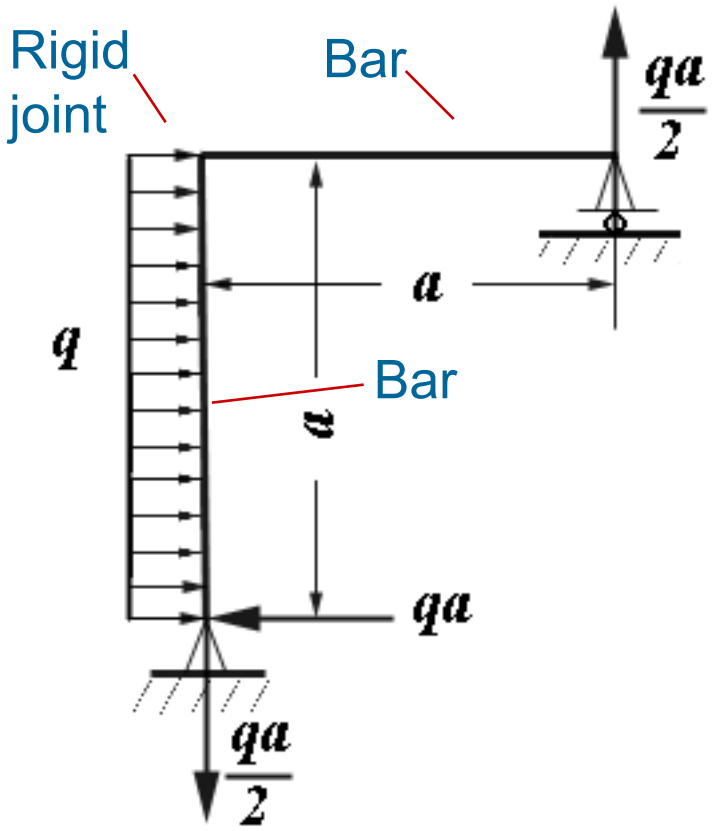


Diagram of Internal Forces for Plane Frames

- Sign convention
  - Axial forces: tension positive / compression negative
  - shear forces and bending moments: the same sign convention when the plane frame is observed from inside.
- Drawing convention
  - Axial forces and shear forces: draw at either side of the bar with sign labeled.
  - Bending moments: draw at *the side under tension* without sign labeled.

Example 12

- Draw the diagram of axial forces, shear forces and bending moments for the plane frame shown below.
- Solution:

1. reaction forces at the supports.

$$\sum M_A = 0 \Rightarrow$$

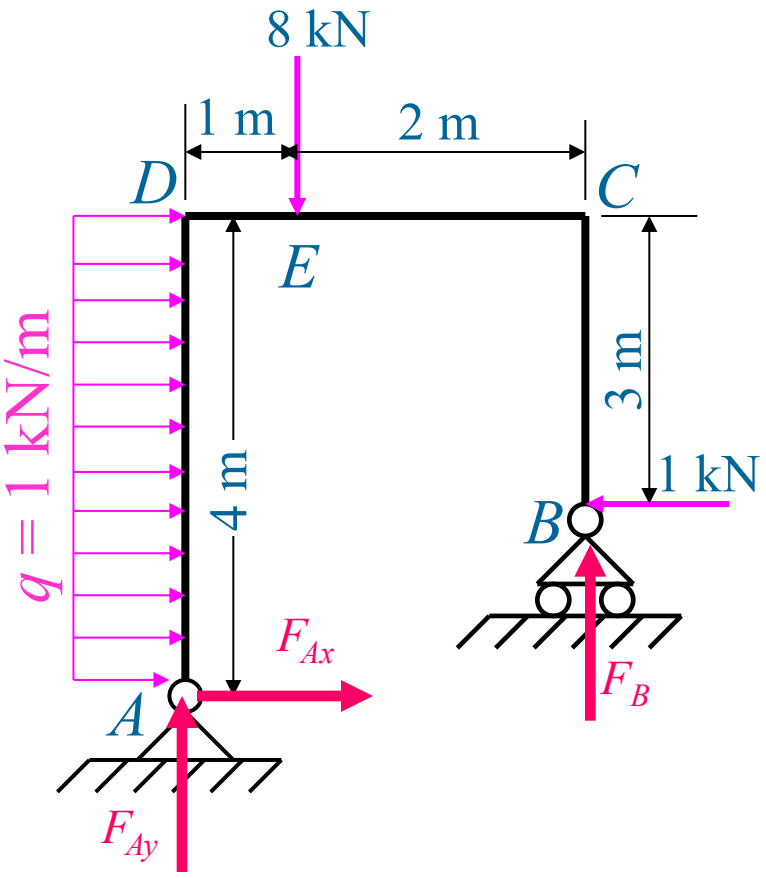
$$F_B = \left(8\times 1 + 1\times 4^2/2 - 1\times 1\right)\Big/ 3 = 5\ \text{kN}$$

$$\sum F_x = 0 \Rightarrow$$

$$1\times 4 + F_{Ax} - 1 = 0 \Rightarrow F_{Ax} = -3\ \text{kN}$$

$$\sum F_y = 0 \Rightarrow F_{Ay} + F_B - 8 = 0$$

$$\Rightarrow F_{Ay} = 3\ \text{kN}$$

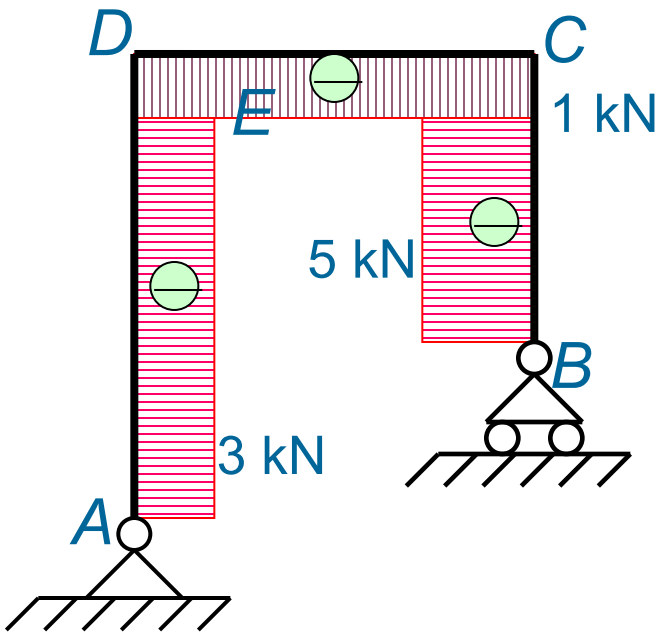


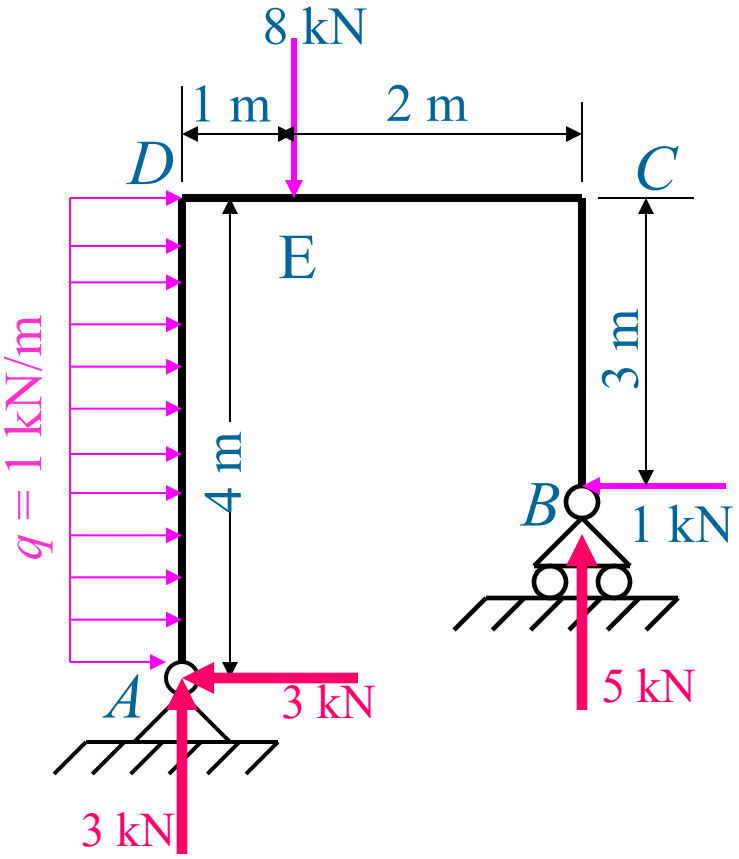
Example 12

2. Diagram of internal forces

1) Diagram of axial forces

$$BC: F_{N1} = -5\ \text{kN}$$
$$DC: F_{N2} = -1\ \text{kN}$$
$$AD: F_{N3} = -3\ \text{kN}$$





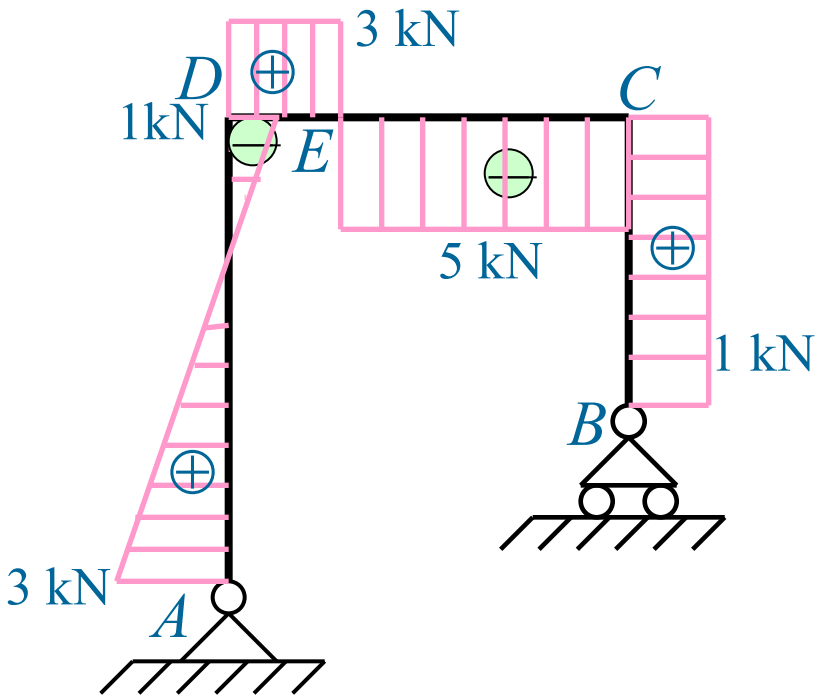
Example 12

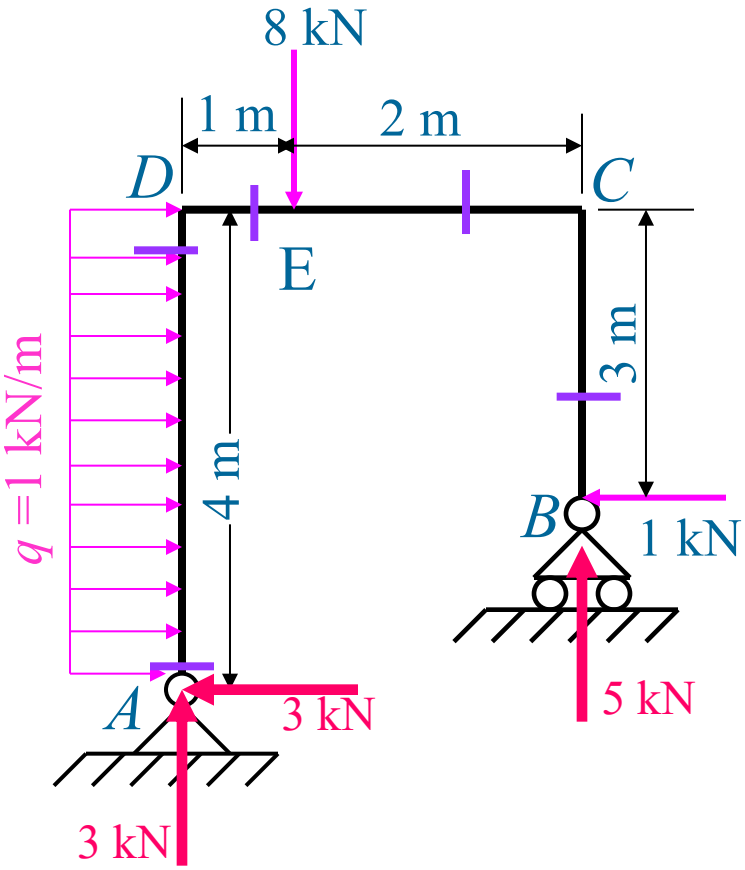
2) Diagram of shear forces

BC: horizontal line

DC: horizontal line

DA: oblique straight line





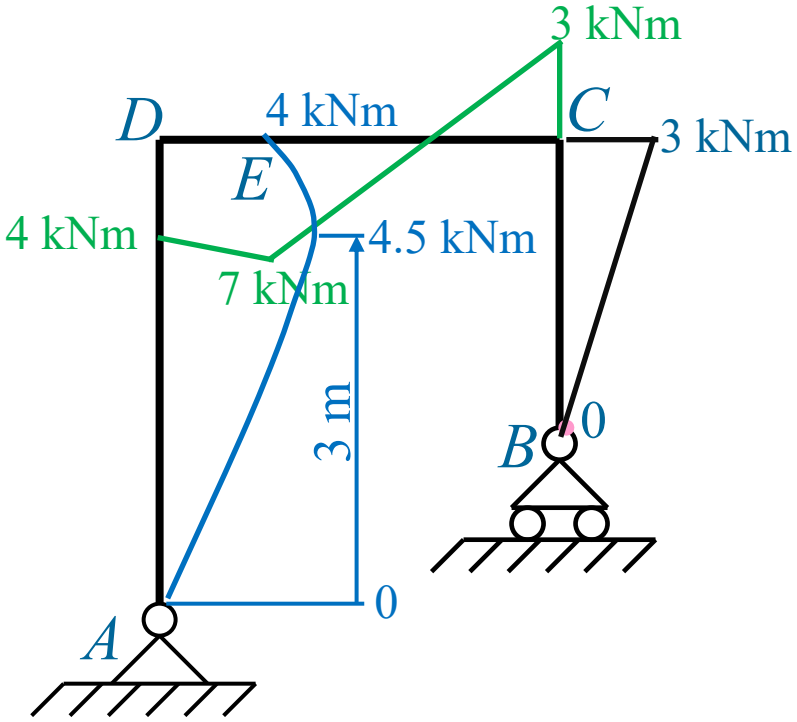
Example 12

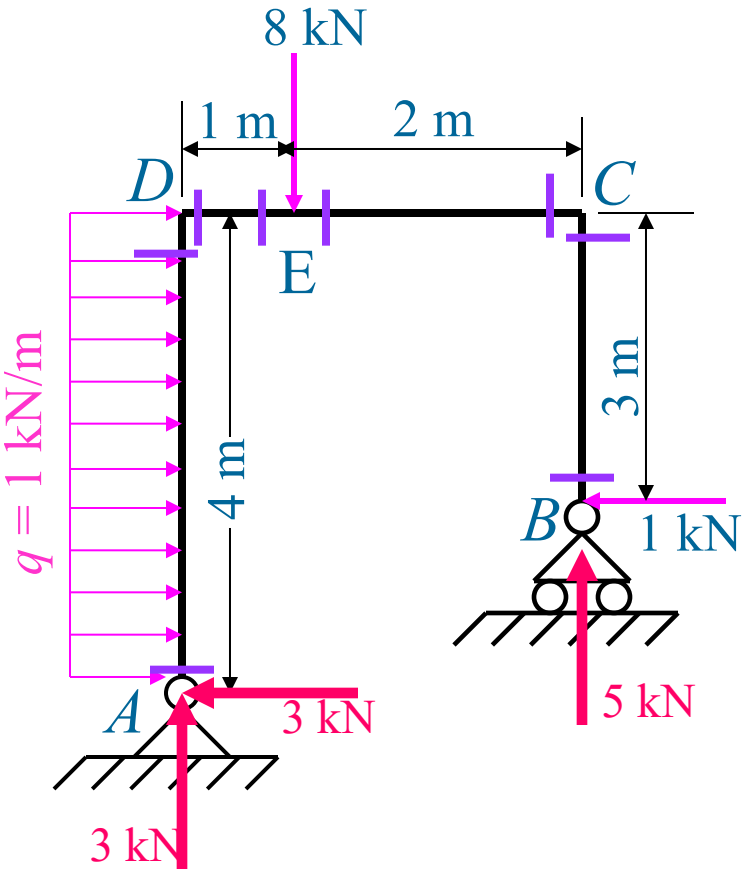
3) Diagram of bending moments

BC: oblique straight line (two-point)

CD: oblique straight line (two-point)

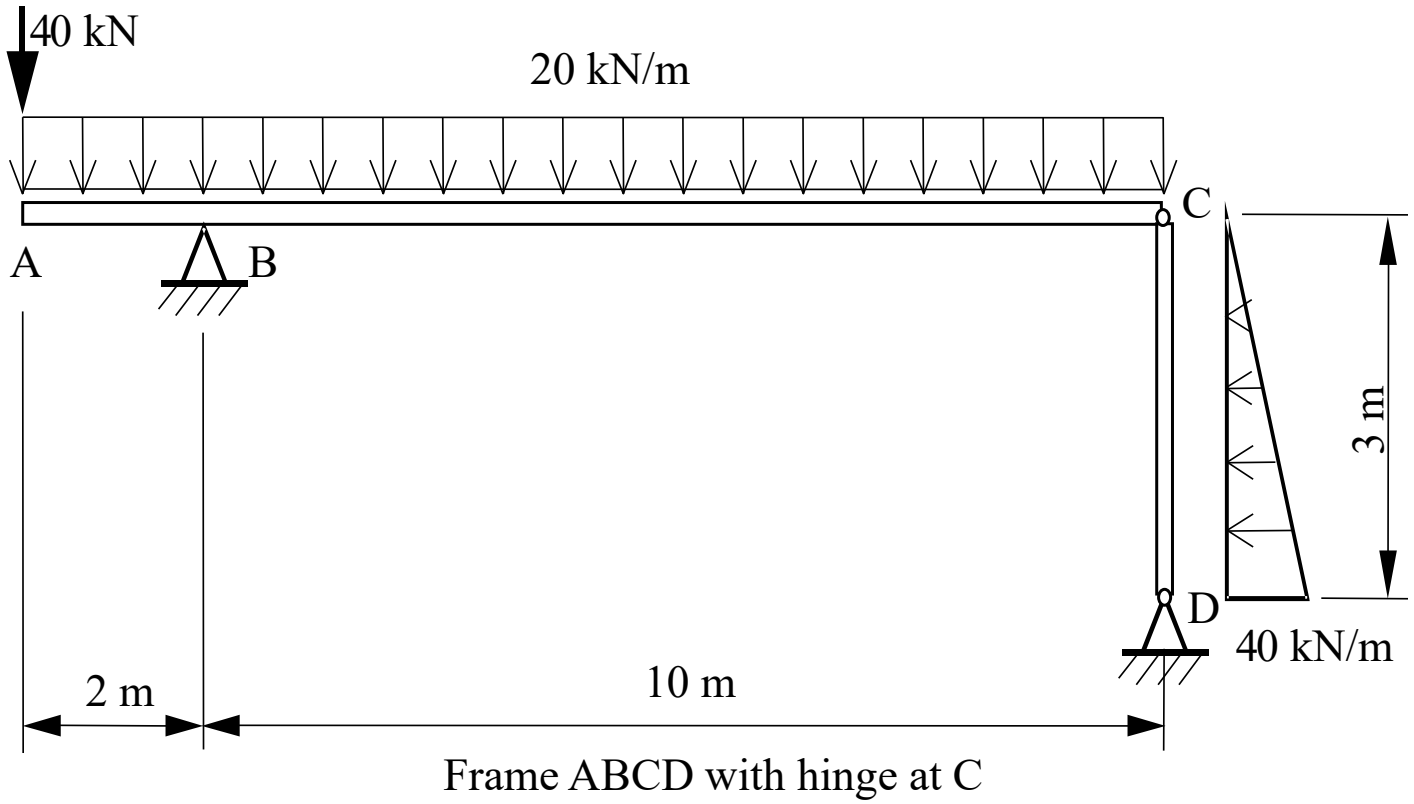
DA: parabola (three-point)



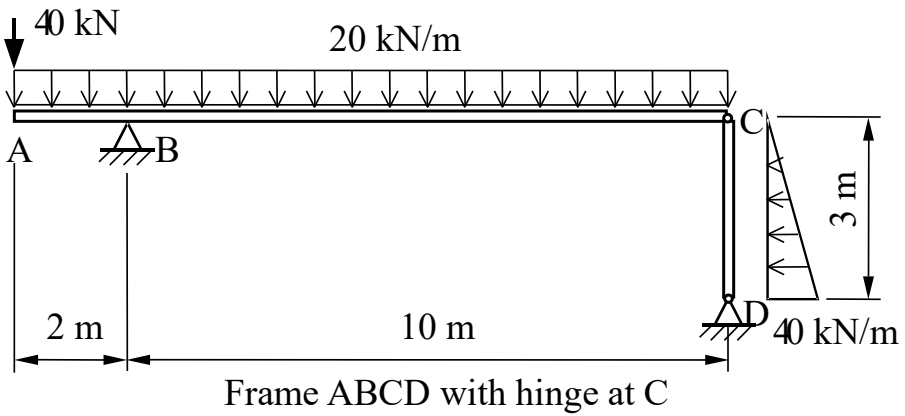


Example 13

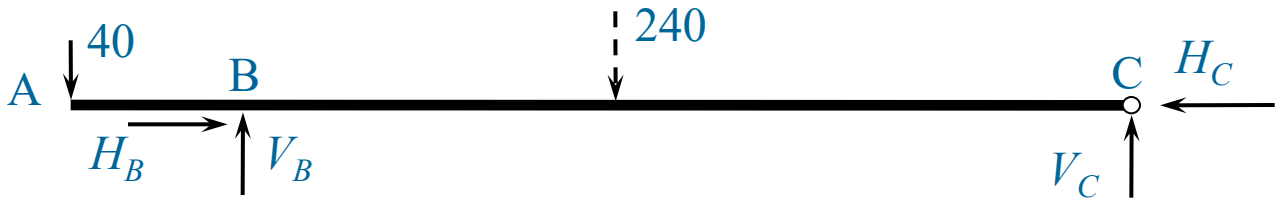
Consider structure ABCD with a hinge at C as shown



Frame ABCD with hinge at C



Frame ABCD with hinge at C



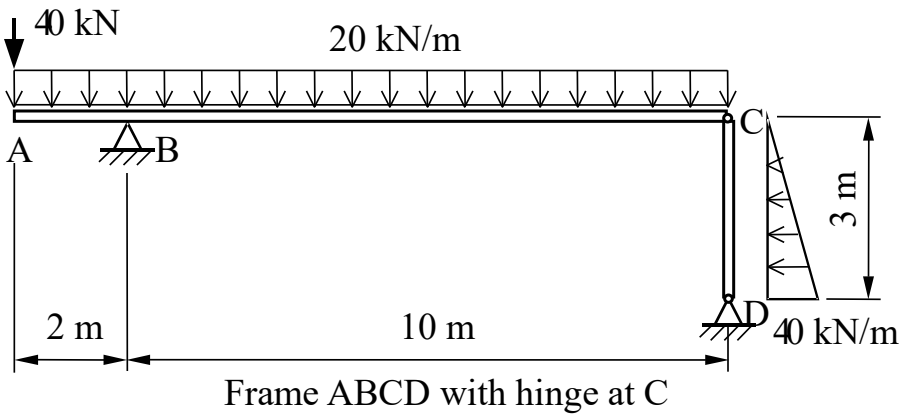
Reactions

- Consider free body ABC. Taking moments about C gives

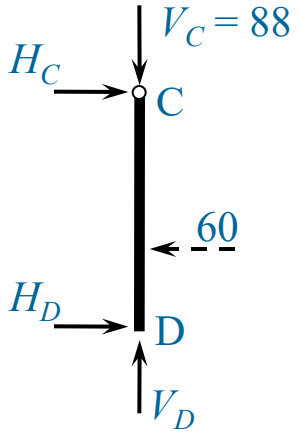
$$V_B \times 10 - 40 \times 12 - 240 \times 6 = 0 \qquad V_B = 192 \text{ kN}$$

- Taking moments about B,

$$-V_C \times 10 + 240 \times 4 - 40 \times 2 = 0 \qquad V_C = 88 \text{ kN}$$



Frame ABCD with hinge at C



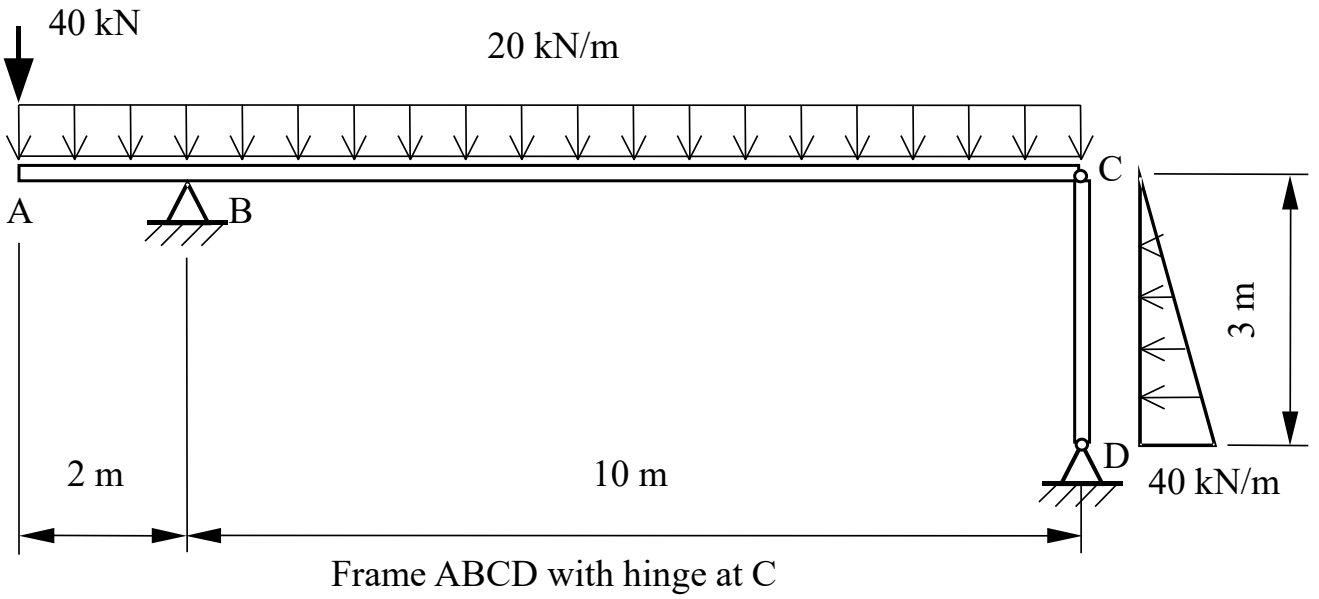
- Then consider free body CD. Taking moments about C gives

$$-H_D \times 3 + \frac{1}{2}(40 \times 3) \times \frac{2}{3} \times 3 = 0 \qquad H_D = 40 \text{ kN}$$

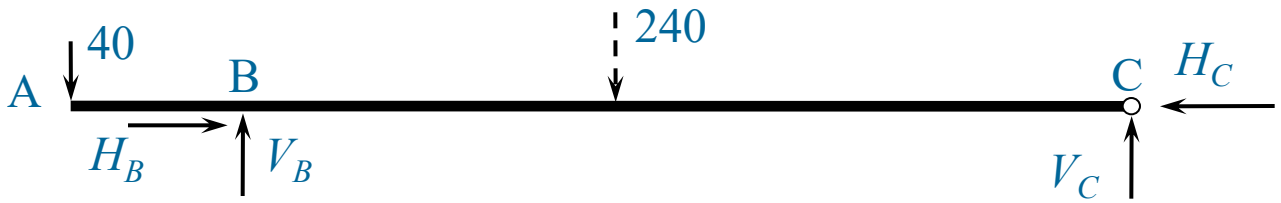
- Consider the equilibrium of forces in the horizontal and vertical directions (x- & y-directions).

(x-dir) $\quad H_C + 40 - 60 = 0 \qquad H_C = 20 \text{ kN}$

(y-dir) $\quad V_D - V_C = 0 \qquad V_D = 88 \text{ kN}$

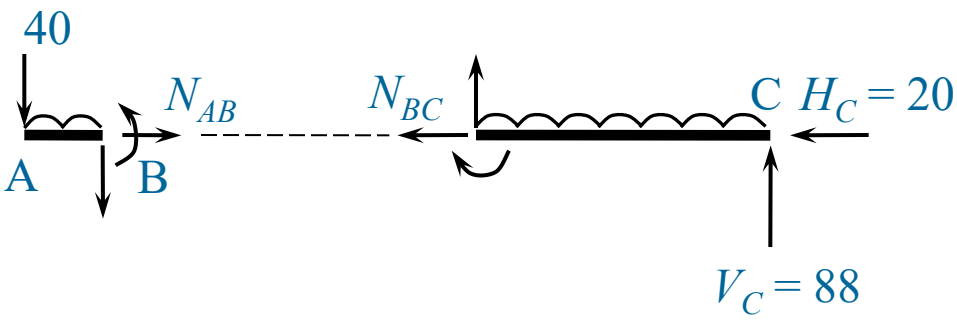


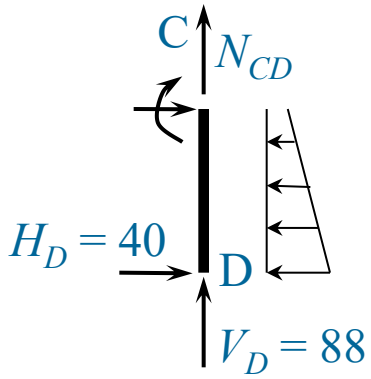
Frame ABCD with hinge at C



- Consider the horizontal equilibrium of free body ABC

$$H_B - H_C = 0 \qquad H_B = 20 \text{ kN}$$



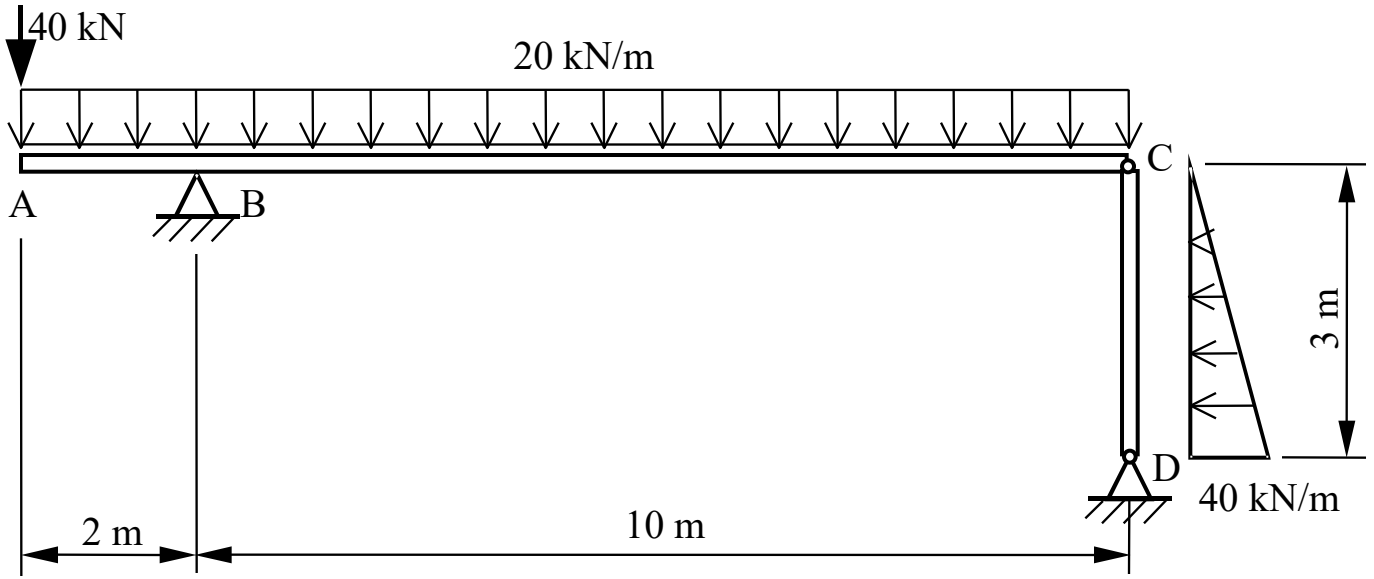


Axial Forces

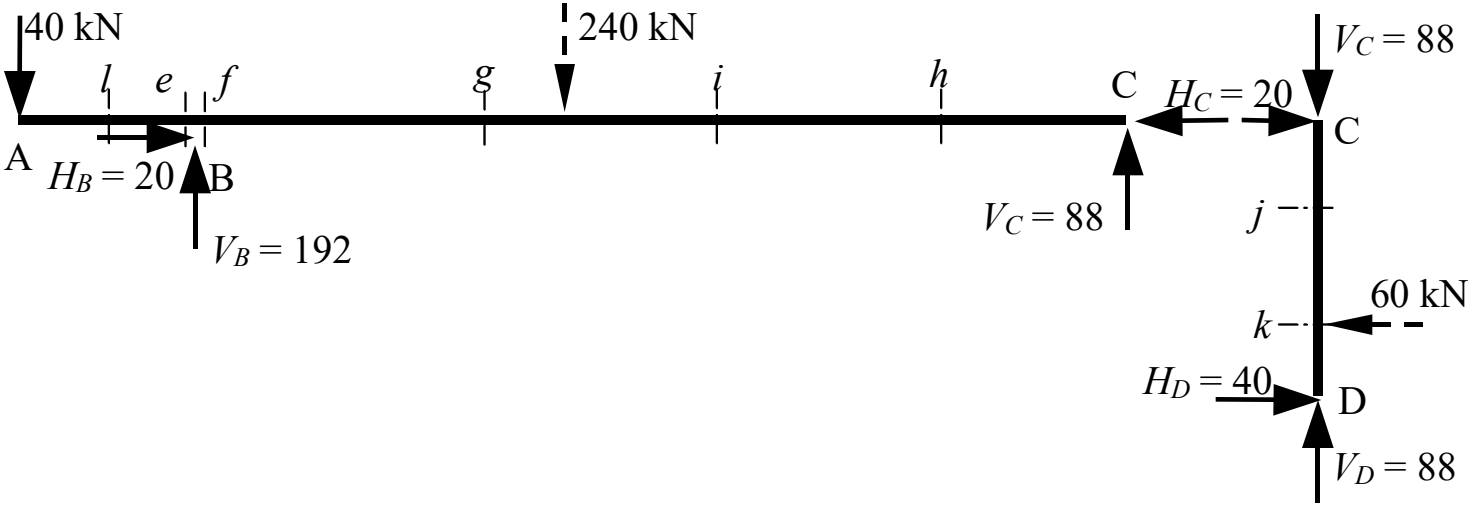
- By inspection of the reactions at B, C and D, one can write

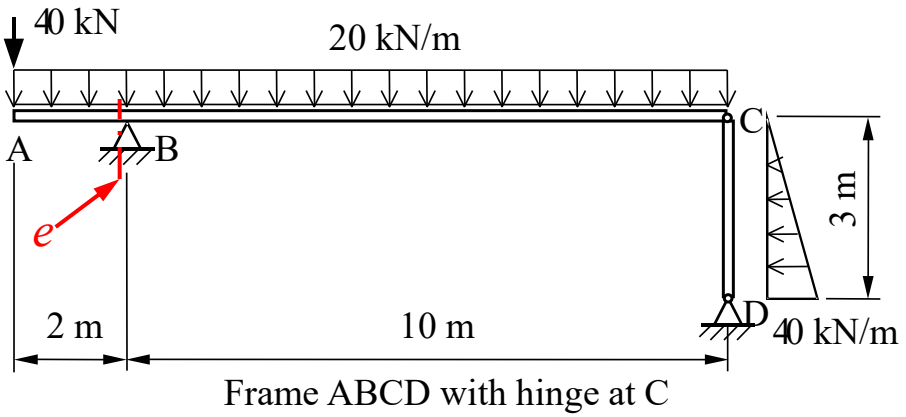
$$N_{BC} = -20\ \text{kN(Compression)} \qquad N_{AB} = 0\,\text{kN}$$

$$N_{CD} = -88\ \text{kN(Compression)}$$



Frame ABCD with hinge at C



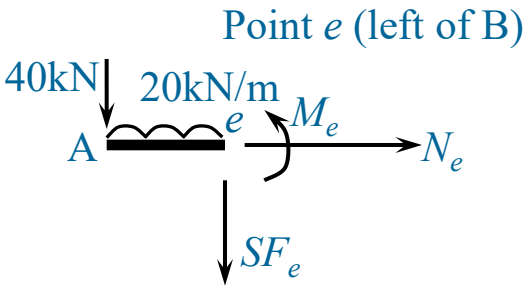


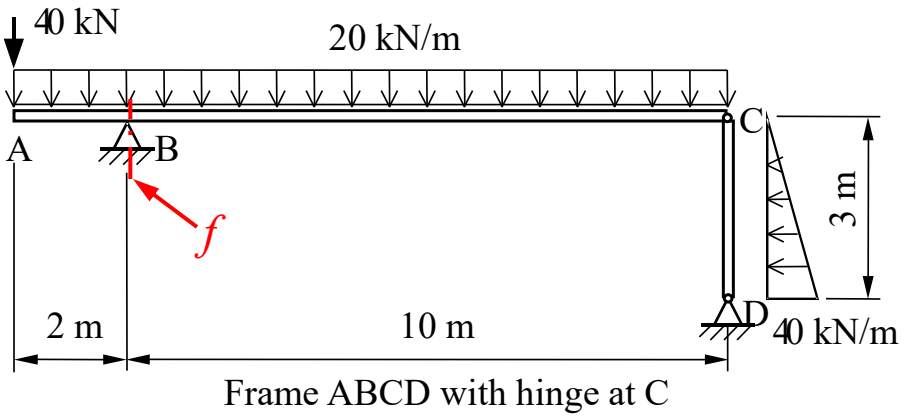
Frame ABCD with hinge at C

Shear forces in member ABC

- The shear force at a section *e* just to the *left of support B* can be obtained from the vertical equilibrium of free body A*e*, i.e.

$$SF_e = -40 - 20 \times 2 = -80\text{kN}$$

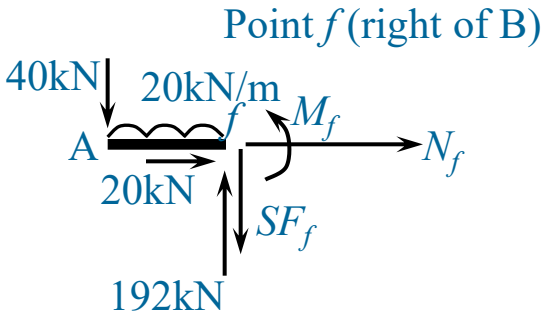


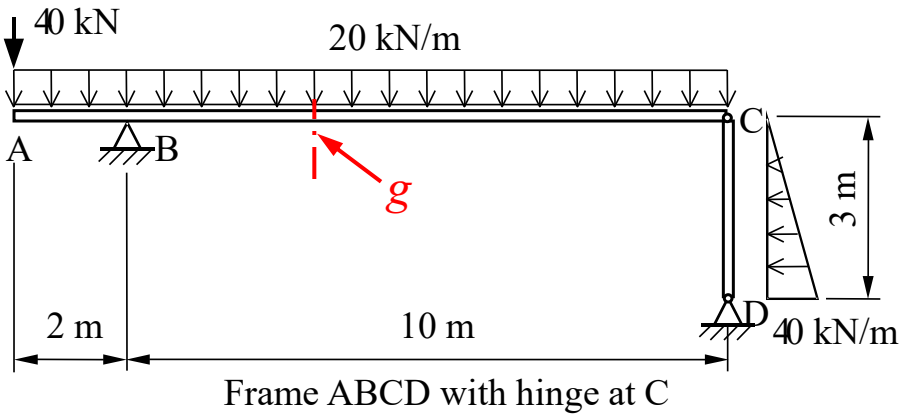


Frame ABCD with hinge at C

- The shear force at a section *f* just to the right of support B can be obtained from the vertical equilibrium of free body A*f*, i.e.

$$SF_f = -40 - 20 \times 2 + 192 = 112\text{kN}$$

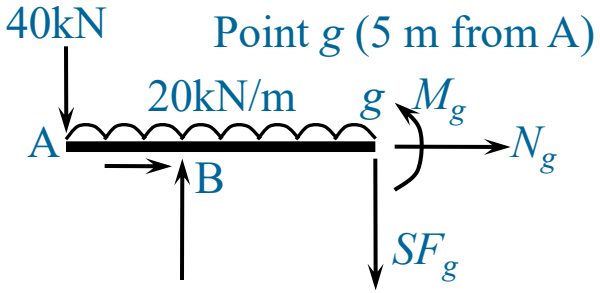


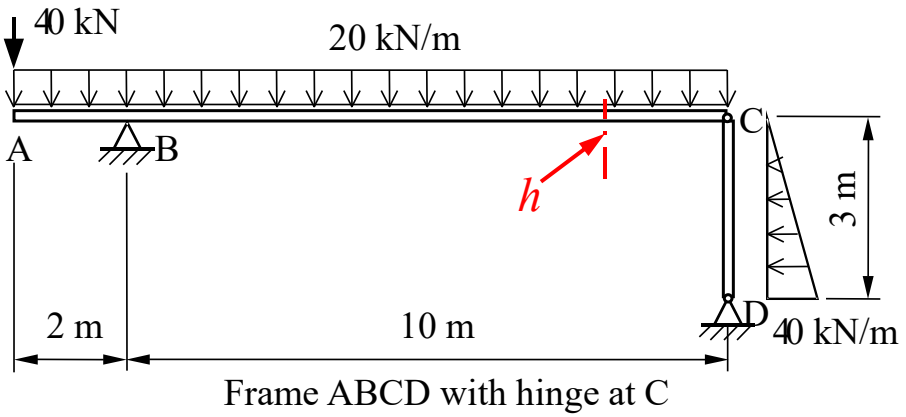


Frame ABCD with hinge at C

- The shear forces at other sections can be obtained similarly. For a section g that is 5m from A, consider free body Ag

$$SF_g = -40 - 20 \times 5 + 192 = 52\text{kN}$$

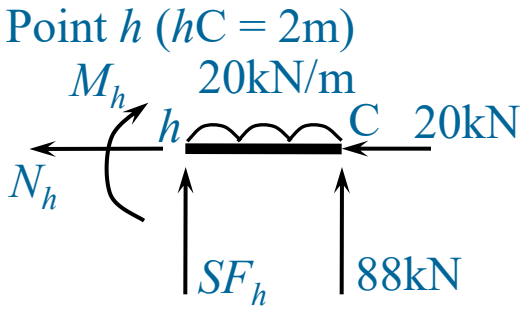


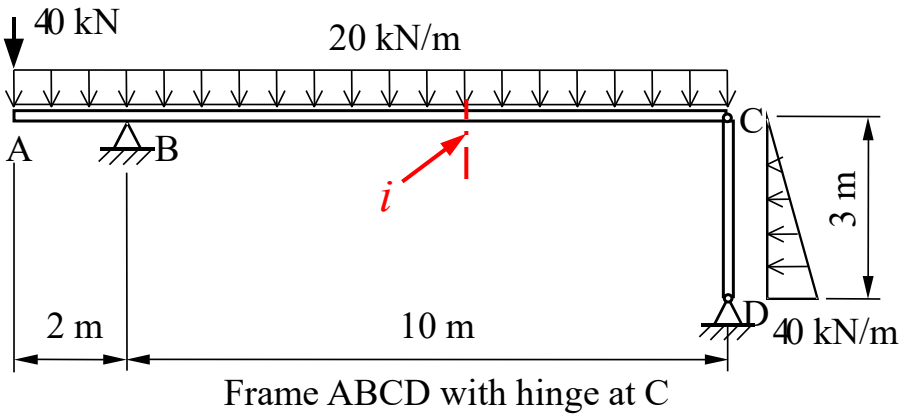


Frame ABCD with hinge at C

- For a section *h* that is 2m from C, consider free body *h*C

$$SF_h = 20 \times 2 - 88 = -48\text{kN}$$

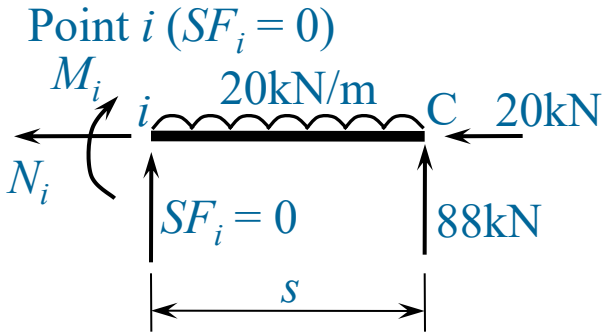


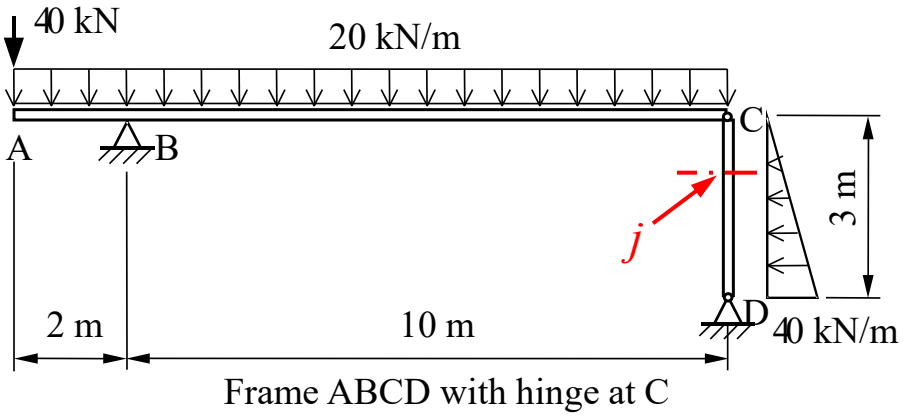


Frame ABCD with hinge at C

- To find the location where the shear force is zero, assume that it occurs at a section *i* at *s* metres from C and consider free body *i*C

$$SF_i = 20 \times s - 88 = 0 \qquad \therefore s = 4.4 \text{ m}$$





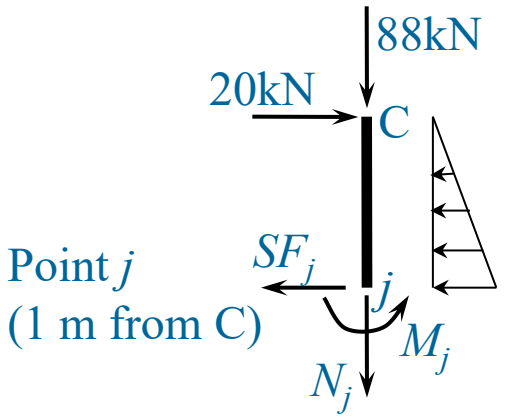
Frame ABCD with hinge at C

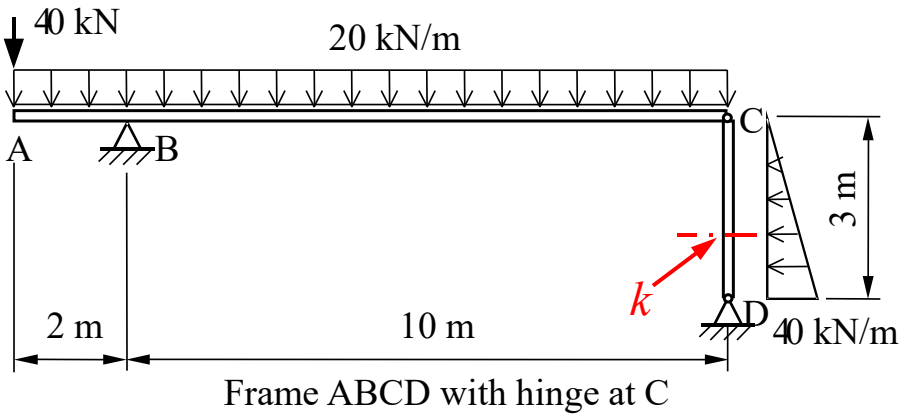
Shear forces in member CD

- The shear force at a section *j* that is 1m from C can be obtained from the horizontal equilibrium of free body C*j*, i.e.

$$SF_j = 20 - \frac{1}{2} \times 1 \times \left(\frac{1}{3} \times 40 \right)$$

$$= 13\frac{1}{3}\ \text{kN} = 13.33\,\text{kN}$$



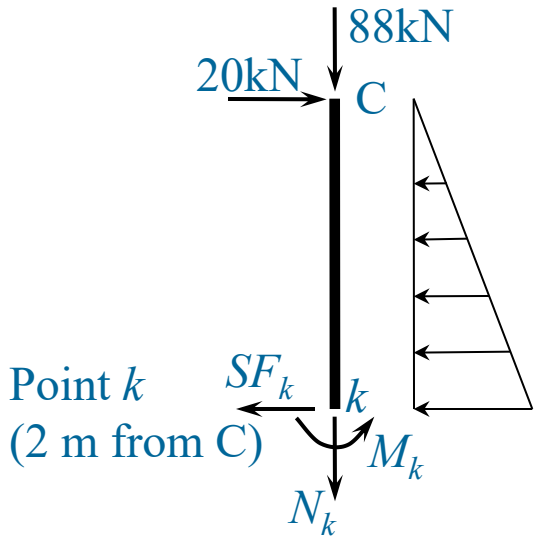


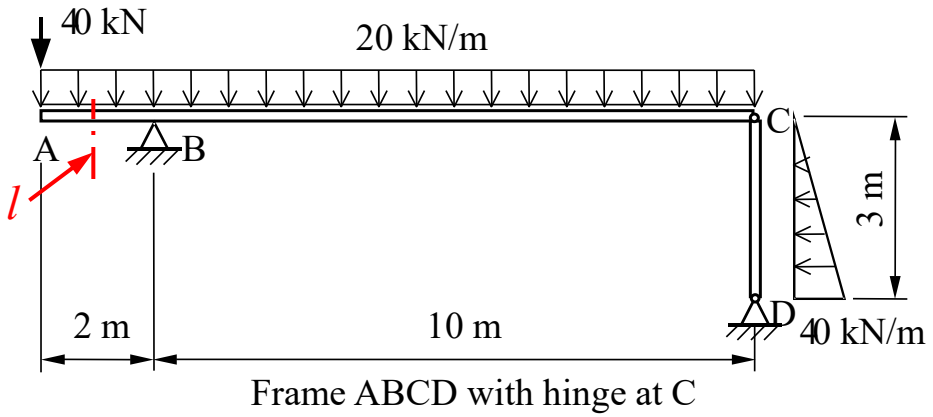
Frame ABCD with hinge at C

- For a section k that is 2m from C, consider free body Ck

$$SF_k = 20 - \frac{1}{2} \times 2 \times \left(\frac{2}{3} \times 40 \right)$$

$$= -6\frac{2}{3}\ \text{kN} = -6.67\ \text{kN}$$



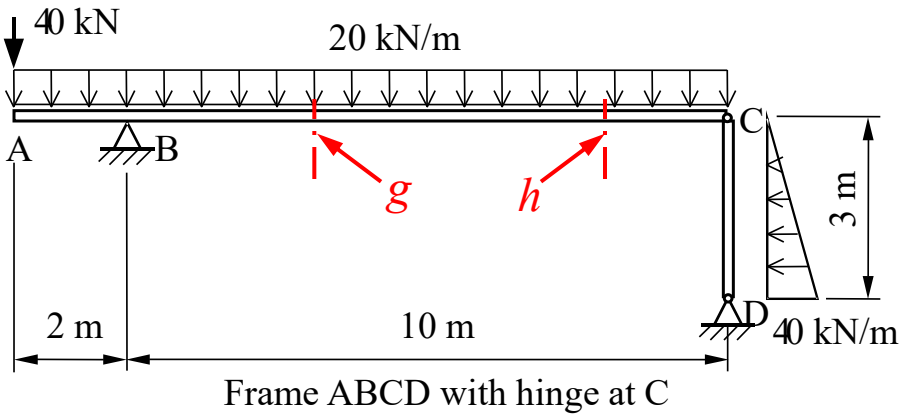


Frame ABCD with hinge at C

Bending moments in member ABC

- The bending moment at a section *l* that is 1m from A can be obtained by taking moments of free body A*l* about *l*, i.e.

$$BM_l = -40 \times 1 - \frac{1}{2} \times 20 \times 1^2 = -50\text{kNm}$$



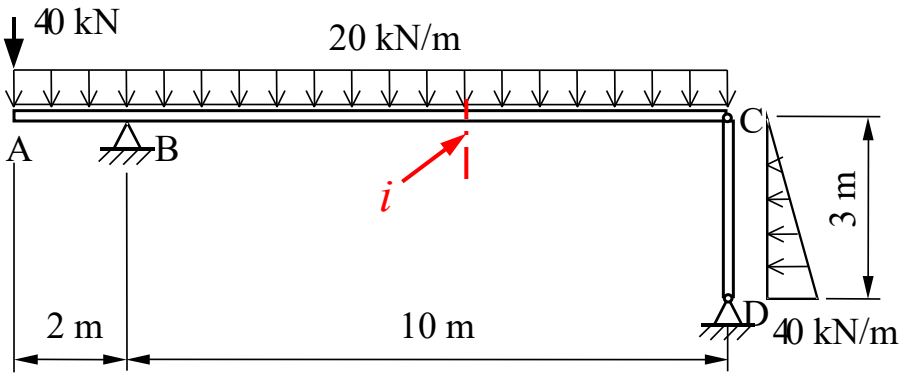
Frame ABCD with hinge at C

- The bending moments at other sections can be obtained similarly.

At B: $BM_B = -40 \times 2 - \frac{1}{2} \times 20 \times 2^2 = -120\text{kNm}$

At g: $BM_g = -40 \times 5 - \frac{1}{2} \times 20 \times 5^2 + 192 \times 3 = 129\text{kNm}$

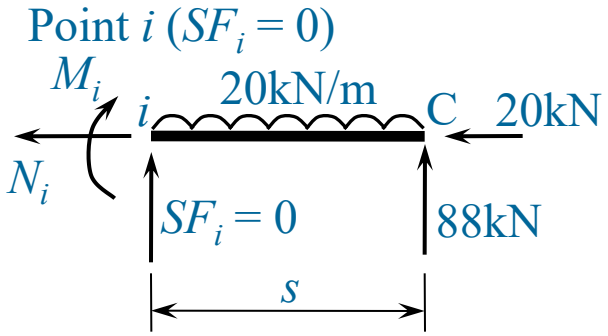
At h: $BM_h = 88 \times 2 - \frac{1}{2} \times 20 \times 2^2 = 136\text{kNm}$

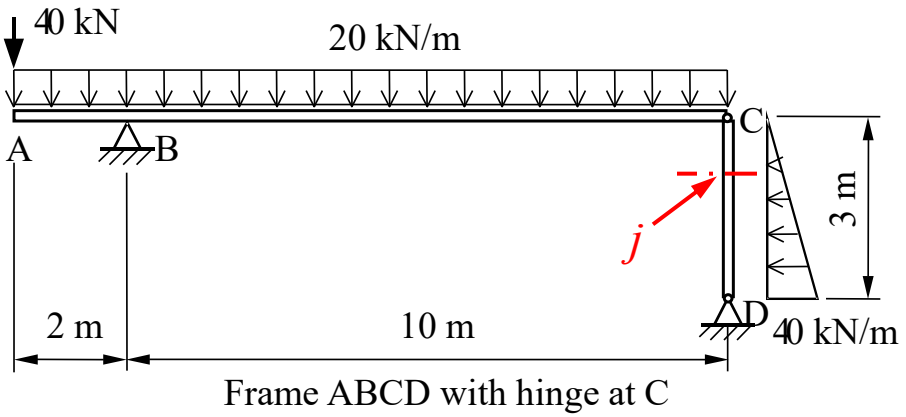


Frame ABCD with hinge at C

- The maximum bending moment normally occurs at the section with zero shear force.

At i:
$$BM_i = 88 \times 4.4 - \frac{1}{2} \times 20 \times 4.4^2 = 193.6\text{kNm}$$

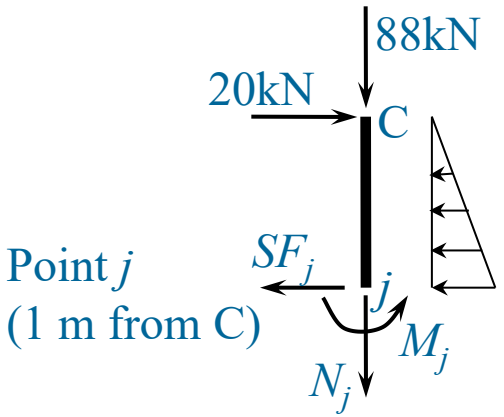


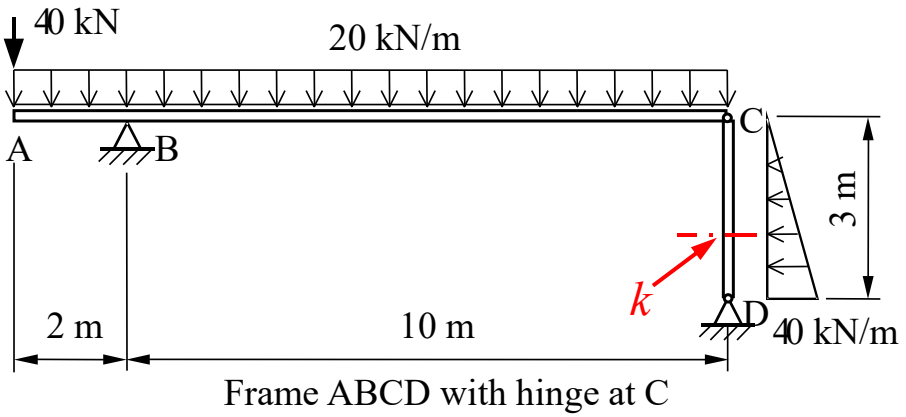


Frame ABCD with hinge at C

Bending moments in member CD

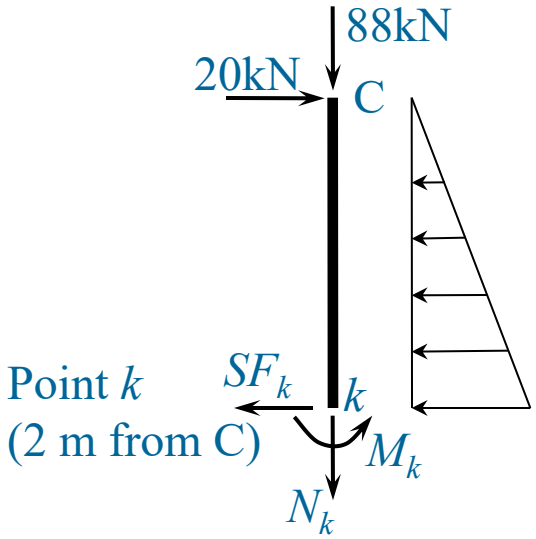
At j: $$BM_j = 20\times 1 - \frac{1}{2}\times 1\times\left(\frac{1}{3}\times 40\right)\times\frac{1}{3} = 17.78\text{kNm}$$



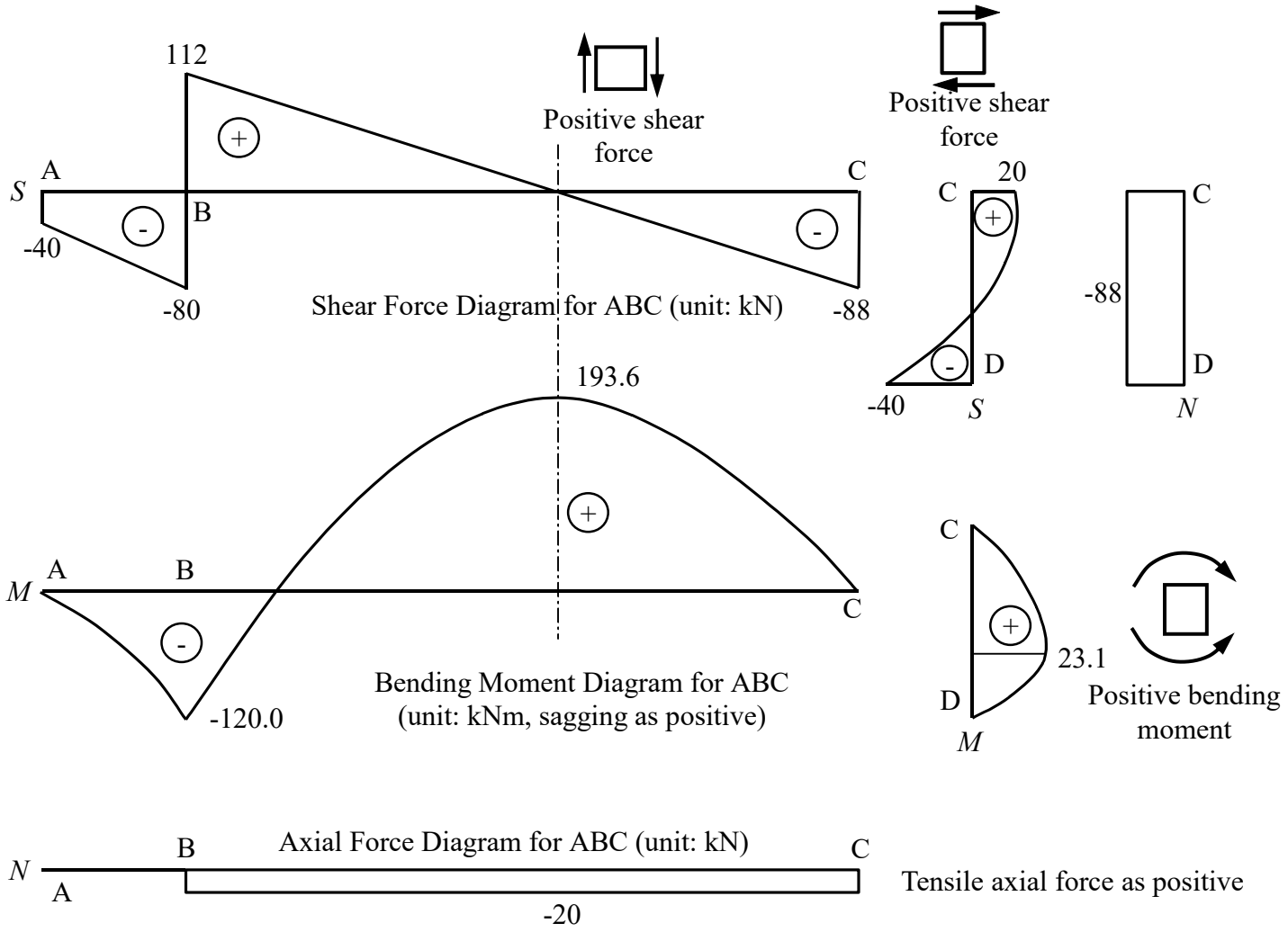


Frame ABCD with hinge at C

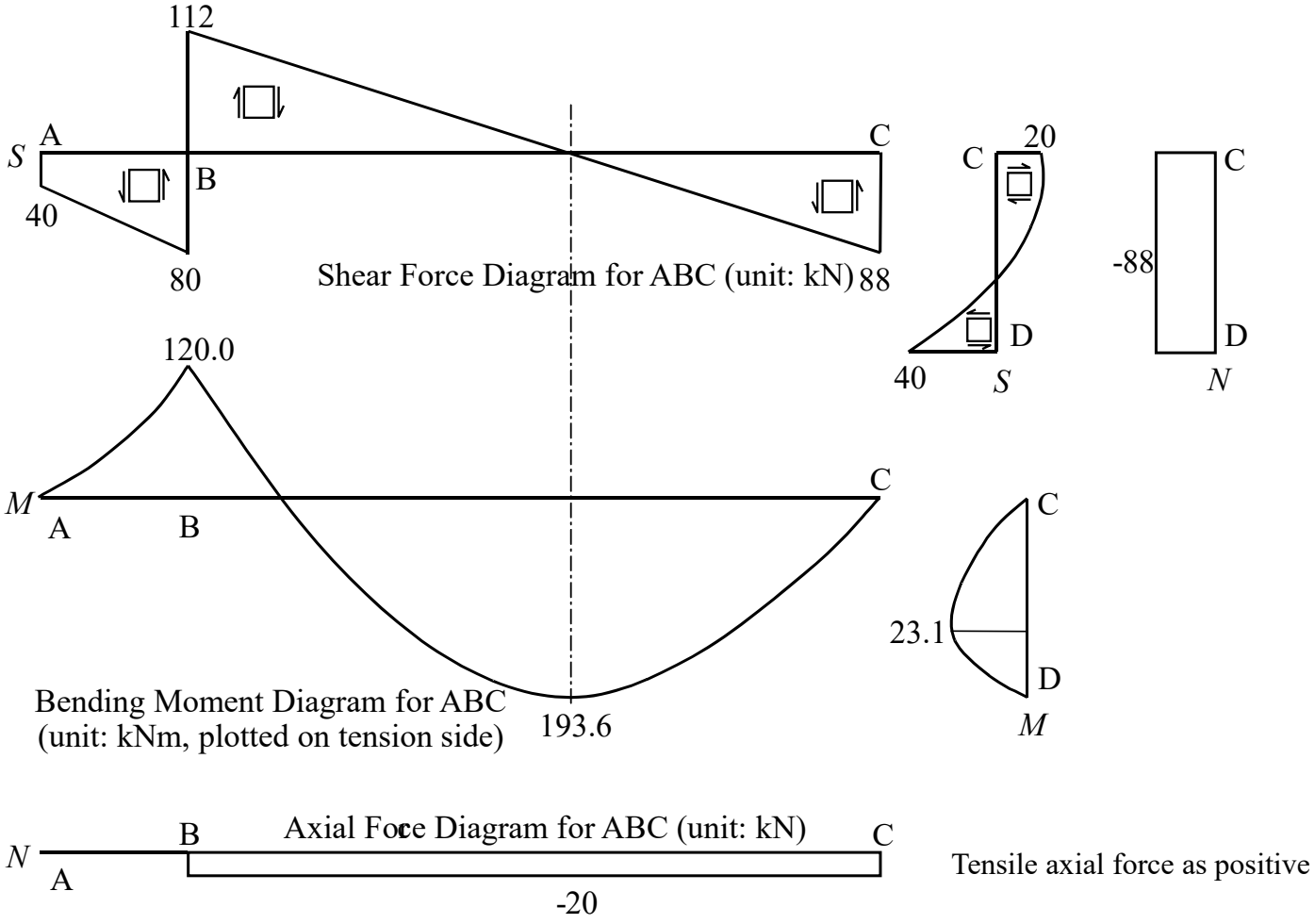
- At k: $BM_k = 20\times 2 - \frac{1}{2}\times 2\times\left(\frac{2}{3}\times 40\right)\times\frac{2}{3} = 22.22\text{kNm}$



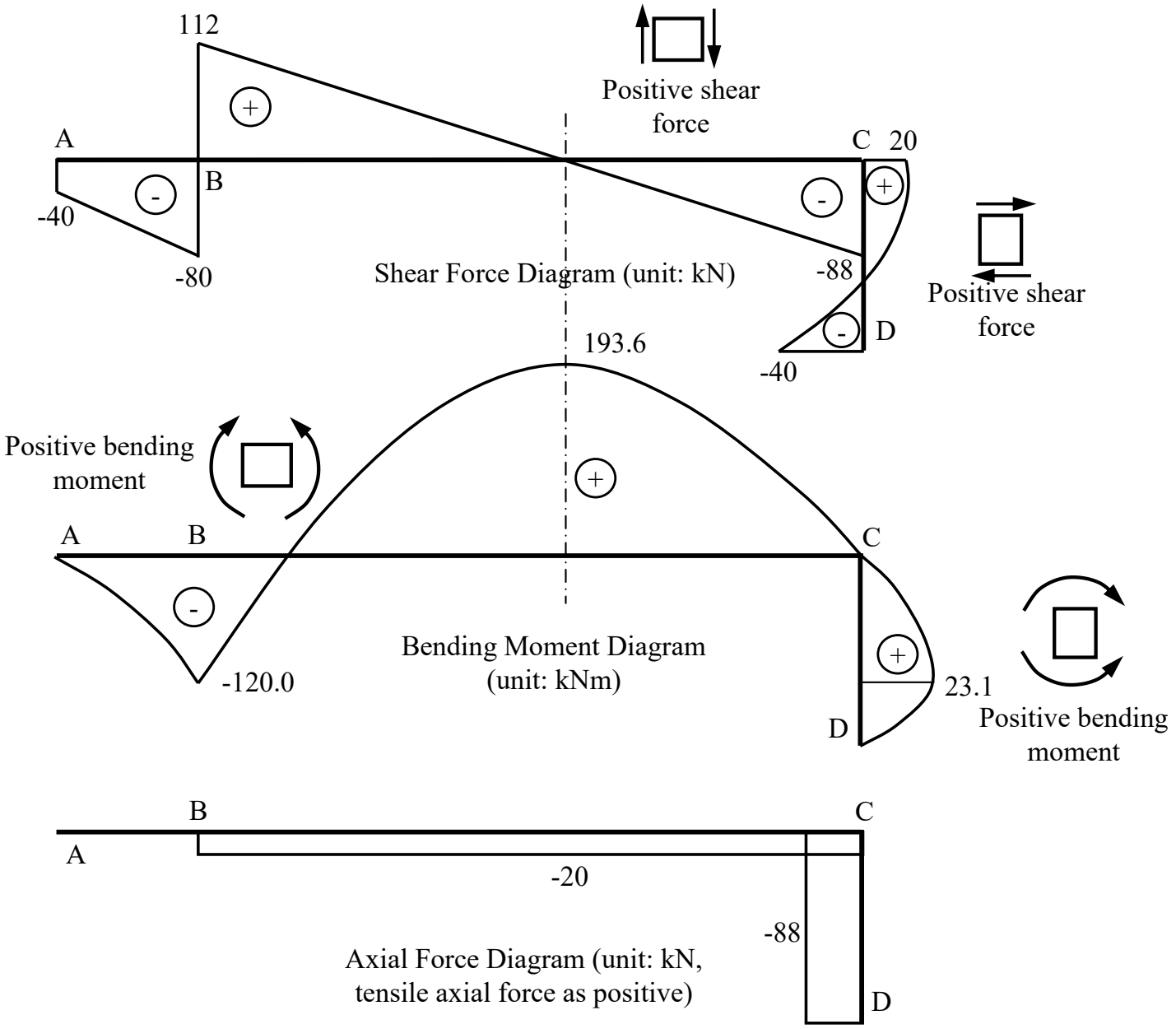
- The bending moment diagrams, shear force diagrams and axial force diagrams can then be plotted with certain sign conventions



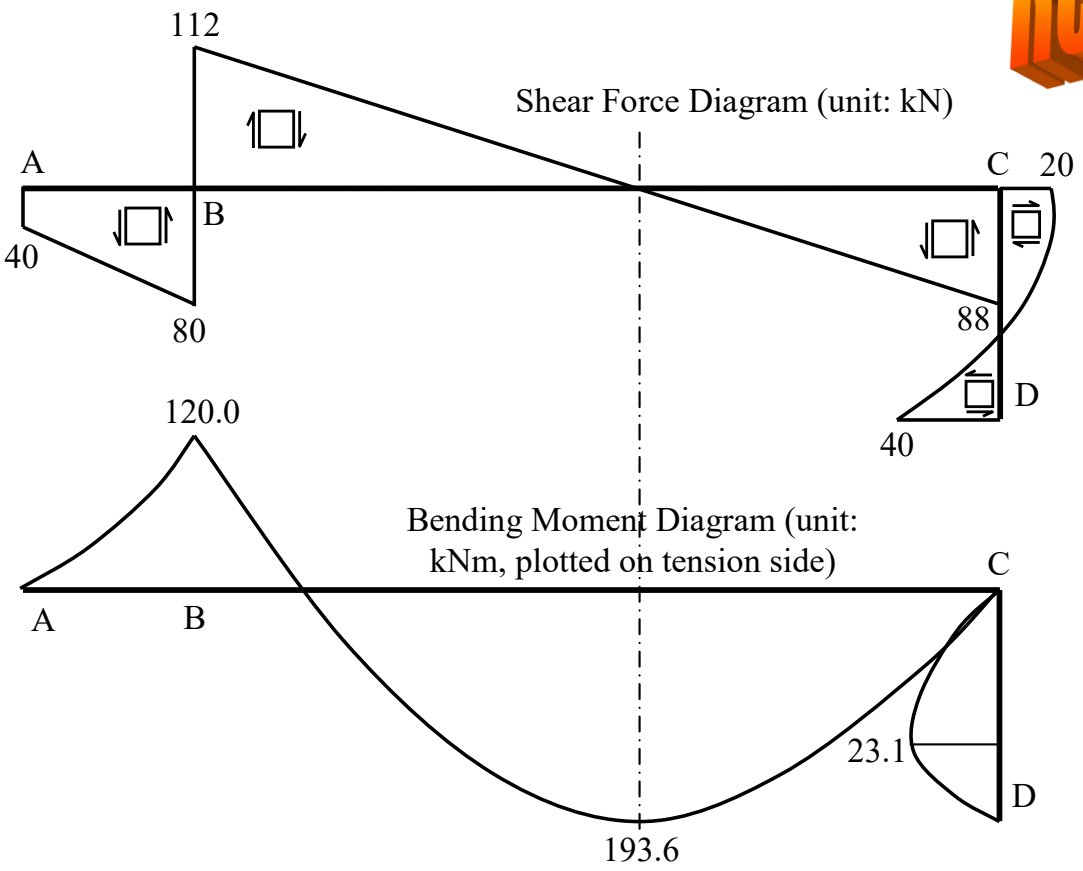
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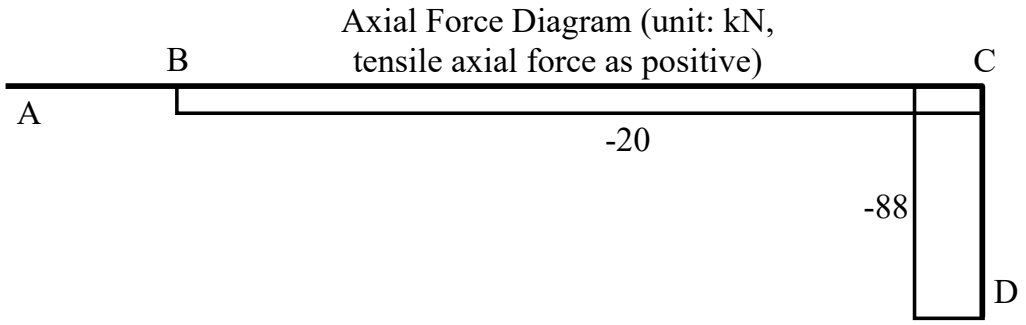


- Diagrams plotted together

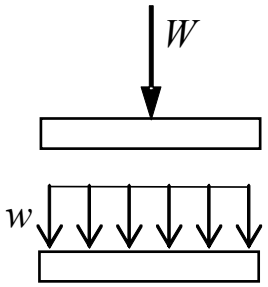


- Diagrams plotted together

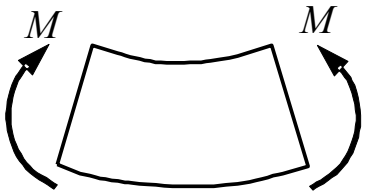




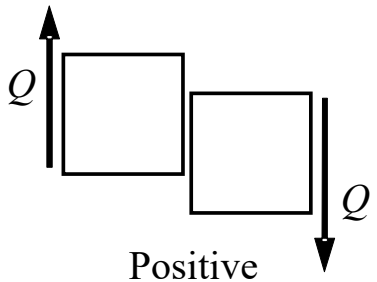
The sign conventions must be observed to avoid confusion



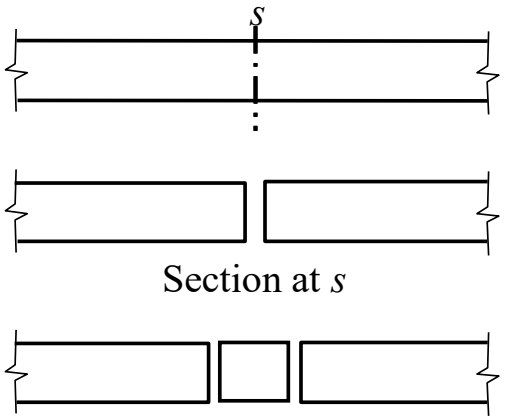
Positive
Loading



Positive
Bending



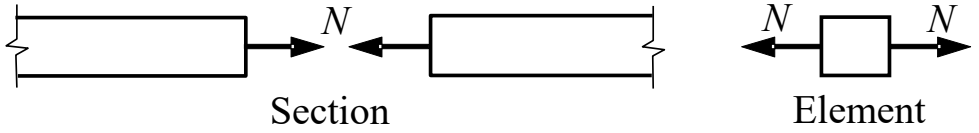
Positive
Shear



Section at *s*

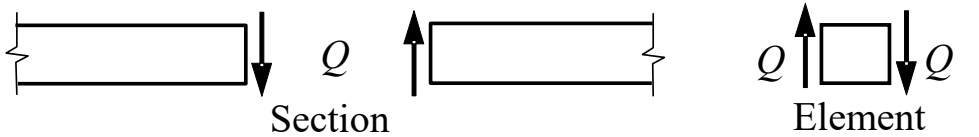
Element at *s*

Two methods of considering point *s*



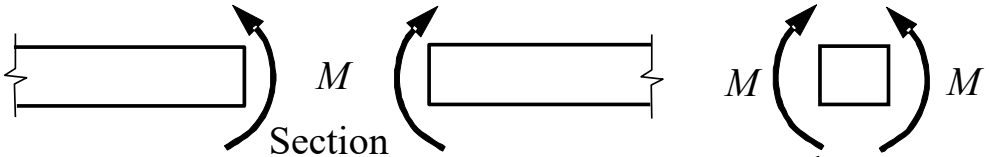
Section Element

Positive axial force



Section Element

Positive shear force



Section Element

Positive bending moment

Notation and Sign Conventions

Summaries for bending moment of beams and frames

- Reviewed the analysis of statically determinate beams and simple frames to find support reactions and internal stress resultants (*N, V*, and *M*), then plotted axial force, shear, and bending-moment diagrams to show the variation of these quantities throughout the structure.
- We considered clamped, sliding, pinned and roller supports, and both concentrated and distributed loadings in assembling models of a variety of structures with different support conditions.
- In some cases, the internal releases were included in the model to represent known locations of zero values of *N, V*, or *M*
- For statically determinate structure, law of equilibrium are adequate to obtain the internal force and reaction force.
- If pin is present, moment is released and moment at that point is 0. Similar apply for axial release and shear release.

THE END